

UNDERGRADUATE FINAL YEAR PROJECT REPORT

Department of Computer Science

IQRA UNIVERSITY



Hydro Mart

Group Number: _____

Batch: 2020-2024

Group Member Names:

Arham ali 18636

Tilal Ahmed 18641

Mirza Humza Saeed 19107

Approved by

.....

Syed Jamal Haider Zaidi

[Assistant Professor]

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Author's Declaration

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Signature and Date

Signature and Date

.....

.....

.....

[Arham Ali]

[Tilal Ahmed]

[Mirza Humza Saeed]

[18636]

[18641]

[19107]

[arham.18636@iqra.edu.pk]

[tilal.18641@iqra.edu.pk]

[mirza.19107@iqra.edu.pk]

Statement of Contributions

The report is made with the contributions of the Undermentioned Students.

1. Arham Ali
2. Tilal Ahmed
3. Mirza Humza Saeed

ABSTRACT

Hydro Mart is a mobile application that seeks to revolutionize the conventional water selling business through a provision of an online platform that links the customers to the local RO Plants. Another advantage of the implemented system is in assisting the customer in straight-forward ordering and payment capabilities, supplier choices, and other general conveniences in making a better purchase. Hydromart has some uniqueness of incorporating the Traveling Salesman Problem (TSP) that enables the delivery riders to determine the shortest possible routes to several delivery points. It is not only fastens the process of delivery, but at the same time, it is a time and resources saver. Hydromart therefore harnesses facilities of the latest technologies like Flutter and Firebase to produce a solution that in the same measure boosts the customer experience and supplier portability. Through the adoption of efficiency, convenience and a sustainable image, the services offered by Hydromart revolutionize the manner in which water delivery and usage is done transforming it into a more efficient and effective business for all stakeholders.

Executive Summary

1. Problem Statement

Definition of the Problem: State the main issue **Hydro mart** project seeks to address in clear and concise terms. This can comprise challenges experienced in conventional water supply methods, and the desirability of a better methodology.

Importance of the Problem: Describe why this problem is important and why there is need to solve this problem. Argue how the problem, if left unresolved, would affect the users, businesses organization and the industry as a whole.

2. Background Information

Historical Context: Describe the historical context concerning the problem. It could involve such areas as historical evolution of water delivery systems and other attempts at integrating it with digitization.

Current Solutions and Limitations: Examine the current approaches or tools utilized to mitigate the problem, analyze the drawbacks or inefficiencies of each approach or tool, and demonstrate where they cannot cater the demanded services or solutions by clients or the industry.

3. Methodology Used to Solve the Problem

Approach Taken: Explain how the problem was addressed or solved in the **Hydro Mart** project. This might entail things such as choosing the right technology (for example, Flutter, Firebase), adopting a user-oriented approach, and reflecting in cycles.

Tools and Technologies: Identify and describe some of the tools, frameworks, and technologies employed within the project, including the Flutter framework for UI, Firebase for backend, and Dart as the language.

Development Process: Explain the various stages, which are research, design, implementation, testing, and deployment of the model.

4. Major Findings

Results Achieved: State the findings made from the study or the conclusion arrived at after the completion of the project. These could be such things as the proper utilization of a convenient mobile application, increased order management, and customer satisfaction.

Key Insights: Talk about any important things you learned while working on the project. This could be how users behave technical problems you ran into, or surprising good points about the tech you picked.

Comparison with Initial Goals: Check how the end result stacks up against the aims and targets you set when you kicked off the project.

5. Conclusions

Summary of the Project: State the overview of the whole project, including the problem statement, the approach taken, and its key results.

Implications of the Findings: Discuss the general implications of the findings for the project, including impacts on the industry that could be involved in any dealings, future research, and further development opportunities.

Future Recommendations: Suggestions on what could be addressed in future, or the potential to improve project, add new features, fine-tune performance, or any other domain to which you can potentially apply this project.

Acknowledgments

On behalf of the three group members, we are able to say thank you so much to our supervisor, Syed Jamal Haider Zaidi for offering his precious time to guide us, suggest and comment throughout the completion of the project. Syed Jamal Haider Zaidi has always encouraged us to work on the tip of our pens, and he has also been kind enough to furnished us with content relevant for the project. We would also like to express our thankfulness to Iqra University for providing this opportunity to execute this valuable project.

Disclaimer

We solemnly declare that research work presented in our UG project, as the case maybe, titled “Hydro Mart” is solely our research work with no significant contribution from any other person or Generative AI Tools. Small contribution wherever taken has been duly acknowledged/cited and that complete project has been written by us in accordance with the latest plagiarism policy declared by HEC and my respective university in-line with the policy for use of Generative AI Tools.

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I undertake that if I am found guilty of any plagiarism in the above titled project/thesis even after award of UG Degree, the University reserves the right to withdraw/revoke my degree and that HEC and the University has the right to publish our name on the HEC/University Website on which names of students are placed who submitted plagiarized project/thesis.

Chapter 1 Introduction

Aim & Objectives

Aim

The overall goal of the **Hydro mart** project is to develop a comprehensive, cross-platform e-commerce application designed especially for hydroponic and water-based products. It will provide smoothness in the process of buying, efficiently managing the inventory, and also providing recommendations to customers by way of personalization.

Objectives

- **Develop a Cross-Platform Application:** Develop Android and desktop applications from a single code base using Flutter.
- **Implement Real-Time Data Management:** Use Firebase for real-time synchronization, authentication, and storage of data.
- **User-Friendly Interface:** There shall be an appropriate user interface for the customer and the supervisor.
- **Scalability and Security:** Make sure that the application scales with the number of users while still maintaining the best security measures for the user.

Relevance of Domain

Importance of E-commerce in Hydroponics

- The hydroponics division is developing exceptionally quick since there's tall request from individuals needing economical and profoundly productive strategies of cultivating. E-commerce stages give an imperative connect between providers and shoppers. They offer comfort, assortment, and competitive estimating for the end-user.

Technological Advancements

- Portable innovation, real-time information administration, are a few of the advancements that have changed the way businesses run their operations and reach out to clients nowadays. Such advances can be utilized within the hydroponics division to create amazing jumps in proficiency and client fulfillment.

Research Question

- How can a mobile application improve the efficiency and reliability of water delivery services in urban areas?
- What impact does the digitization of water delivery services have on customer satisfaction and operational efficiency?
- How effective are route optimization algorithms, like the Traveling Salesman Problem, in reducing delivery times and fuel consumption for water delivery services?
- What are the security implications of handling personal data in mobile applications for essential services like water delivery, and how can these be mitigated?

Report Structure

Project Idea Pertaining to Industry

Industry Relevance The focus of our project is towards meeting the hydroponics industry's needs, which is a niche market in very rapid growth. Hydro Mart aims to fill this gap by offering a professional e-commerce platform that draws suppliers and consumers closer and gives a solution that is tailored to the unique needs of this industry.

Industry Challenges

- **Supply Chain Management:** Efficiently managing inventory and supply chains.
- **Customer Reach:** Expanding market reach to attract more customers.
- **Personalization:** Offering personalized recommendations to increase sales.

Project Contribution

Hydro Mart will help with inventory management, expand to a multi-platform presence to reach more consumers, and leverages machine learning to give every customer a curated shopping experience. These are some of the big challenges of the industry.

Market Value/Need

Market Demand

An e-commerce business for hydroponics products is a niche but has immense potential due to new trends towards environmentally friendly production of food. This business plan envisions an e-commerce site or app that allows potential customers easy access to all the products they need to start their own 'green' farm.

Potential for Growth

Investment with **Hydro Mart** will be worthwhile over the coming years as with the exponentially increasing demand of hydroponics in near future, it is going to be the perfect time for **Hydro Mart** to generate revenue as its highly scalable architecture will lead it to grow with the market and with user and transaction capacity.

Chapter 2 Background And Technology

Introduction

In this chapter, we discuss the technologies that we have selected for the implementation of the Hydro Mart project. We start by discussing the historical background, the problem domain, compare the problem with other similar solutions in the market and describe normatively the problem and the solution. We provide engineers with related research papers in subsequent section and describe the software technologies that were adopted during the built of this project.

Flutter

Detail Background of the Technology

Flutter is an open-source User Interface software development kit alias SDK. It was developed by Google and announced in 2015, then released officially, the support of Google, in December 2018. This assists a developer in designing from one codebase fully high-quality, natively compiled applications for all platforms: Android, iOS, Linux, macOS, Windows, Google Fuchsia, and the web. This multi-platformability cuts development costs and timelines by a large margin. Flutter is written in the programming language Dart, also developed by Google. It is an object-oriented language whose syntax resembles JavaScript but is built to build fast and scalable web and mobile applications. It is, intrinsically, widgets; hence, it's versatile and flexible enough to be whatever you can conceive of. Everything in a Flutter app, from layout to buttons to text, is just widgets. This makes for a consistent user experience across different platforms.

Software Technologies Specification to Be Used in the Project

- **Flutter SDK:** The main toolkit used for developing the user interface (UI) of the **Hydro Mart** application. It provides the tools necessary to build, test, and deploy the application across various platforms.
- **Dart:** The programming language utilized within the Flutter framework. Dart is essential for implementing the business logic and core functionalities of the application.

Firebase

Detail Background of the Technology

Firebase is a platform developed by Google for creating mobile and web applications. Originally founded in 2011 as a real-time database service, Firebase was acquired by Google in 2014 and has since expanded to include a variety of tools and services aimed at simplifying app development. These services include a real-time NoSQL database, authentication, analytics, cloud storage, hosting, and more. Firebase offers seamless integration with other Google services, making it a powerful choice for developers seeking a robust backend solution.

Discussion of Problem Domain with Historical Background

Historically, building backend services for mobile apps required significant effort, resources, and infrastructure management. Developers had to set up and maintain servers, manage databases, handle authentication, and ensure data security. This complexity often led to increased development time and costs.

Firebase addresses these challenges by offering a comprehensive suite of backend services, collectively known as Backend-as-a-Service (BaaS). Firebase simplifies the development process by handling server-side functionality, allowing developers to focus on frontend development and user experience. This shift has enabled faster development cycles and more efficient resource allocation.

Comparison of Similar Solutions with Your Proposed Solution

- **AWS Amplify:** AWS Amplify is another popular BaaS platform offering similar backend services. While both Firebase and AWS Amplify provide robust tools for app development, Firebase's deep integration with other Google services (such as Google Analytics, Google Ads, and the Google Cloud Platform) and its ease of use make it a preferred choice for many developers, especially those already invested in the Google ecosystem.

- **Microsoft Azure Mobile Apps:** Microsoft Azure Mobile Apps offers backend services similar to Firebase, including data storage, authentication, and real-time notifications. However, Firebase's real-time database capabilities, broad community support, and user-friendly interface give it an edge over Azure Mobile Apps, particularly for developers looking for a more streamlined solution.
- **Summary of 4-5 Research Papers Related to the Technology**
 1. **"Real-time Data Synchronization with Firebase":** The paper showcases how the real-time database of Firebase allows quick synchronization of data across different devices so that users could access information from anywhere in an updated form. This paper points out the advantages associated with handling data in real-time in applications where instant updates are critical..
 2. **"Security and Privacy in Firebase Applications":** The following research work has surveyed the security features of Firebase, together with best practices toward its data privacy applications. This paper has succeeded in covering subjects revolving around data encryption, user authentication, and access control, with a focus on the importance of adhering to the security protocols when using BaaS platforms.
 3. **"Firebase vs. Traditional Backend Services: A Performance Comparison":** In this paper, we will focus on the performance, scalability, and ease of use of Firebase in comparison with traditional backend services. This study highlights real time processing abilities of Firebase and its subsequent impact on the ease with which mobile applications are developed hence reducing time-to-market.
 4. **"The Role of Backend-as-a-Service (BaaS) in Modern App Development":** The purpose of this paper is to evaluate the influence of BaaS platforms, such as Firebase, on recent application design. It elaborates on how BaaS has revolutionized developers' approach to backend development making resource management more effective and deployed applications be faster

5. Software Technologies Specification to Be Used in the Project

- **Firestore Authentication:** Used for managing user authentication in the **Hydro Mart** application. Firestore Authentication provides a secure and easy-to-implement solution for user sign-up, sign-in, and account management, supporting various authentication methods including email/password and social media logins.
- **Firestore Cloud Firestore:** A scalable and flexible cloud database used to store and sync data for the **Hydro Mart** application. Cloud Firestore allows for real-time data synchronization across all clients, ensuring that users always have the most current information.
- **Firestore Cloud Storage:** This acts as a media file storage for user-uploaded files like profile pictures and CNIC images. Firestore Cloud Storage is designed with high security, great performance, and ease of use, hence an excellent choice in relation to large volumes of user credentials.

Chapter 3

Requirements & Methodology

Problem-Solving Approach

Planning

For the **Hydro Mart** project, we used a cross-platform mobile app developed in Flutter. Another benefit of this method is, that it gives you wide coverage on all supported environments; Backend Services for Firebase Real-time data synchronization, user authentication and storage. This will be feature for things like personalized recommendations and better inventory management.

Steps to Achieve the Solution

1. **Requirement Gathering:** Understanding user needs through surveys and feedback.
2. **Design:** Creating wireframes and UI/UX designs.
3. **Development:** Building the application using Flutter and integrating Firebase.
4. **Testing:** Conducting unit, integration, and user acceptance testing.
5. **Deployment:** Publishing the application on multiple platforms.
6. **Maintenance:** Regular updates and user support.

Implementation Details

Logic

The application will use Flutter for the frontend, allowing a single codebase for Android applications. The backend will leverage Firebase for real-time database management, user authentication, and storage. Machine learning models will be used for recommendation systems and inventory forecasting.

Constraints

- **Cross-Platform Compatibility:** Ensuring the app works seamlessly on all supported platforms.
- **Real-time Synchronization:** Maintaining consistent and real-time data updates.
- **Scalability:** Ensuring the system can handle increasing numbers of users and transactions.

Platform

- **Frontend:** Flutter (Dart)
- **Backend:** Firebase (Firestore, Authentication, Storage)

Strategy for Implementation

1. **Modular Development:** Breaking down the project into smaller, manageable modules.
2. **Agile Methodology:** Using iterative development cycles with regular feedback.
3. **Continuous Integration/Continuous Deployment (CI/CD):** Using GitHub Actions for automated testing and deployment.

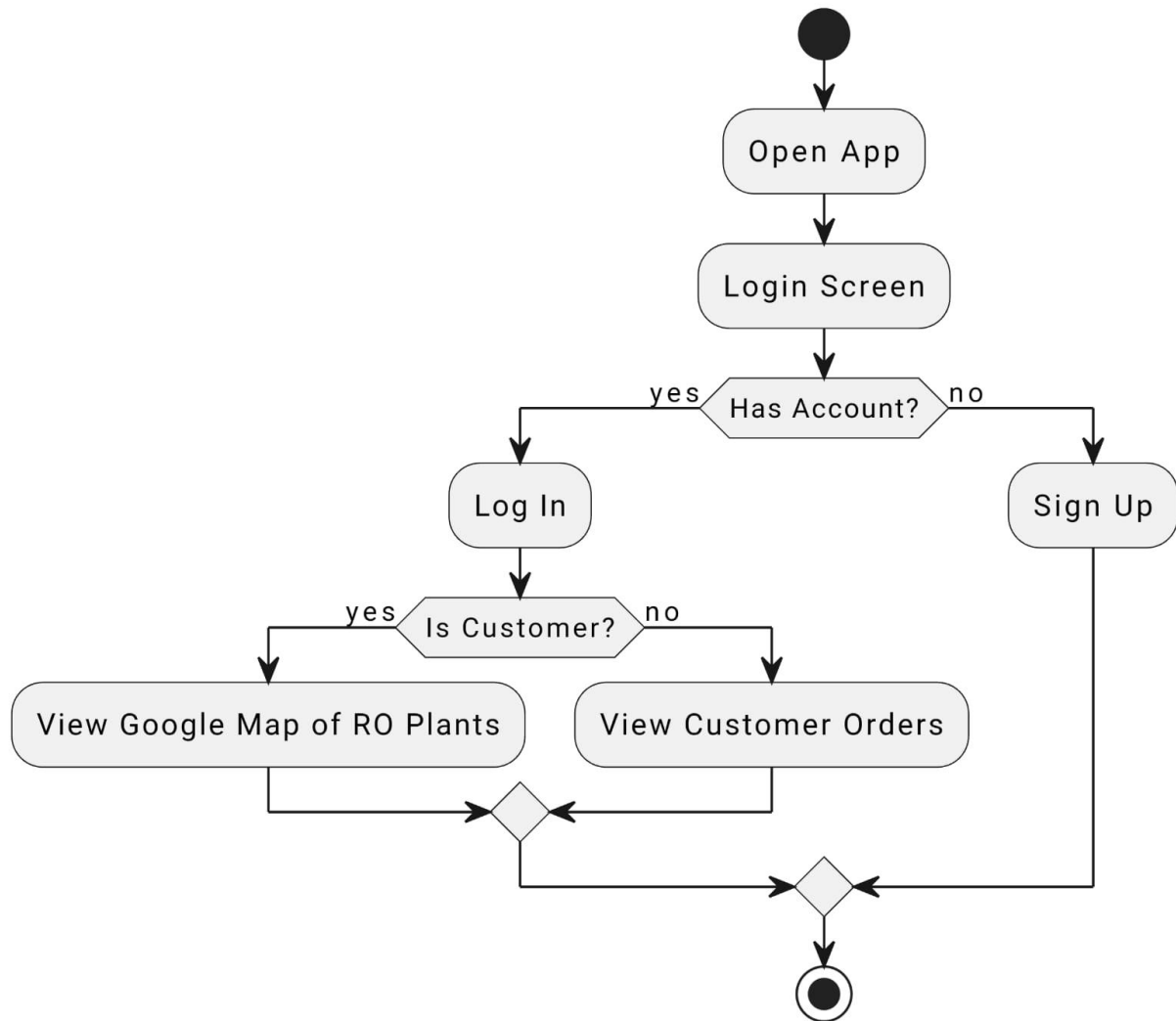
3. Justification for Chosen Approaches

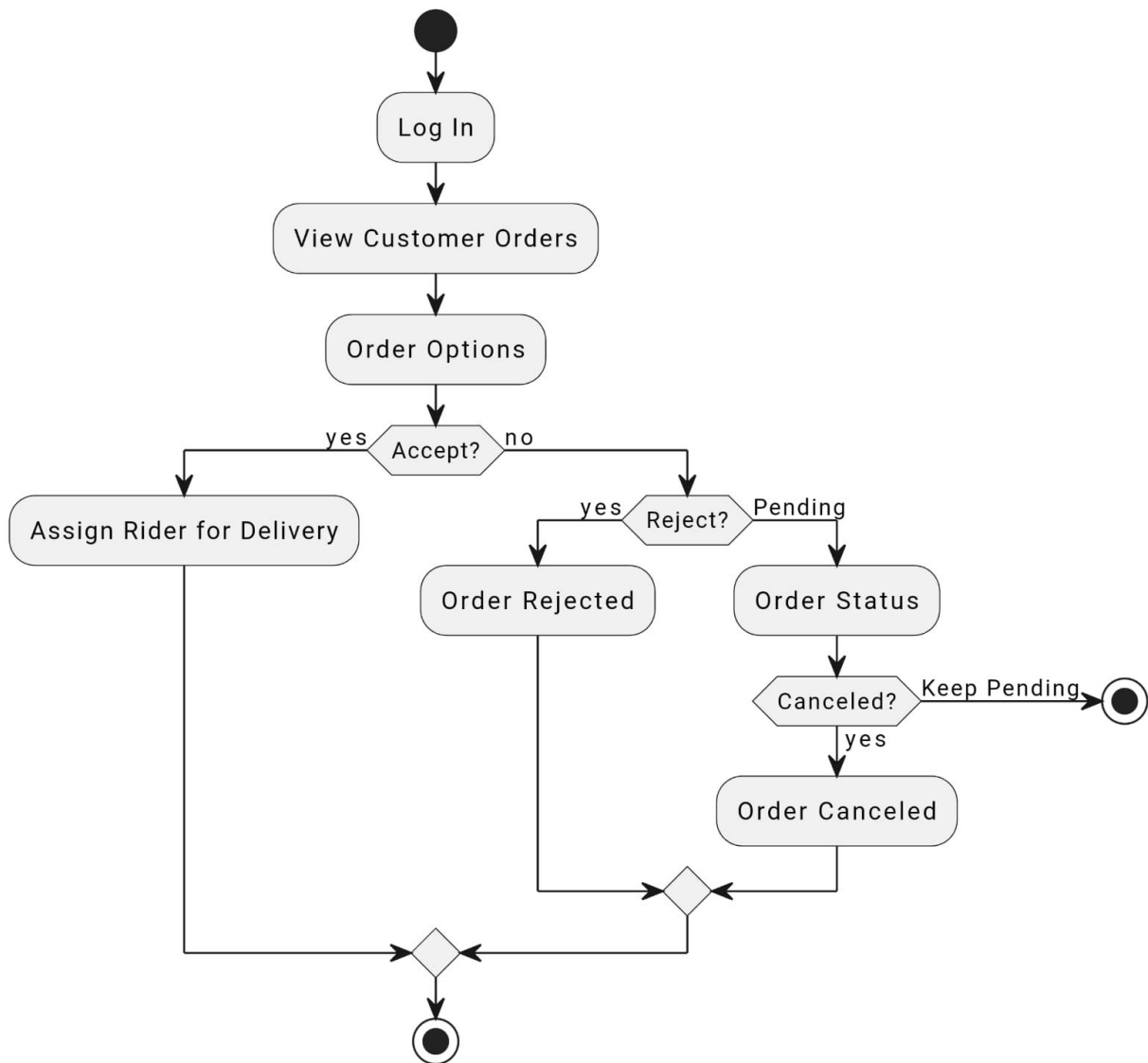
Why You Selected the Specific Strategies, Platforms, or Languages

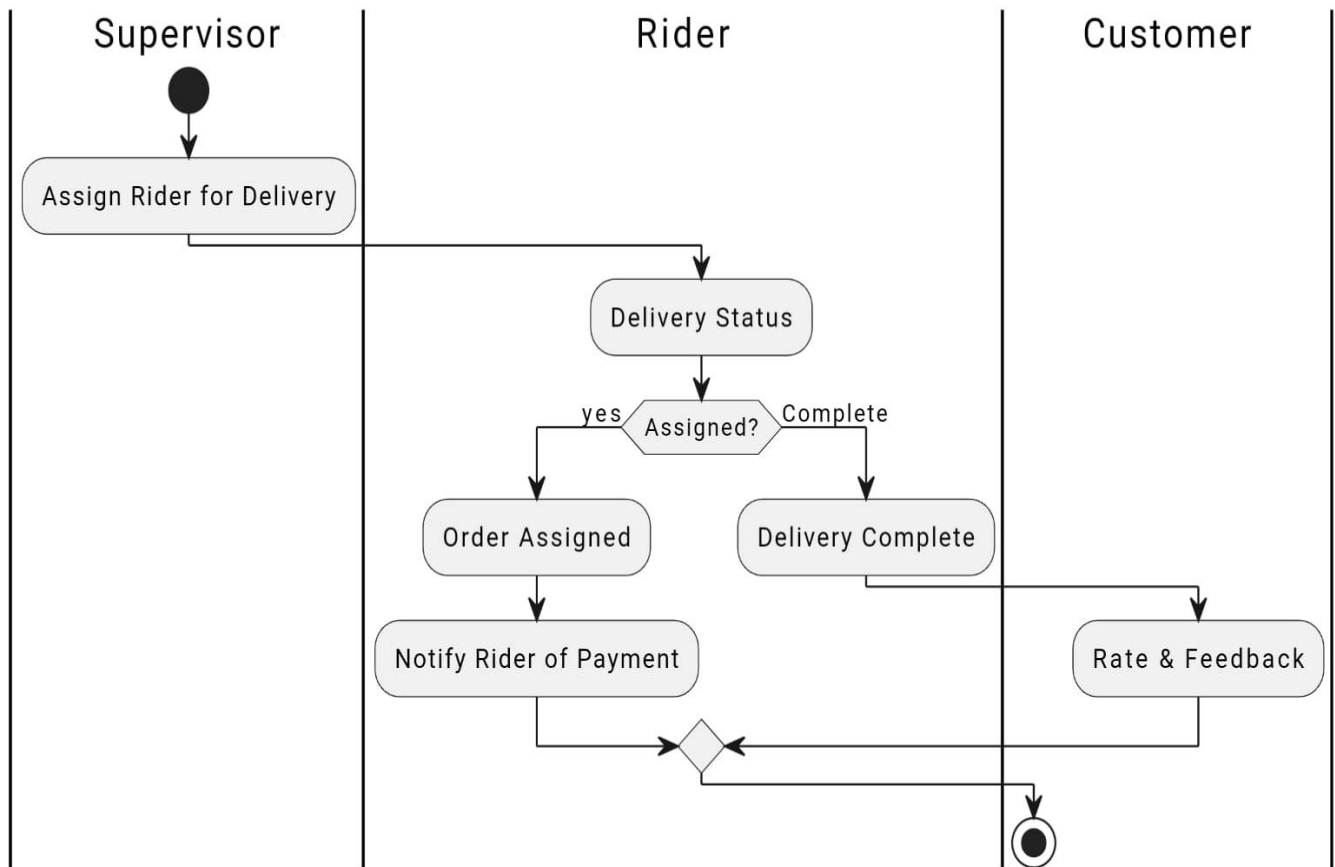
- **Flutter:** Offers a unified codebase for multiple platforms, reducing development time and effort.
- **Firebase:** Provides a comprehensive backend solution with real-time database, authentication, and storage, reducing the need for custom backend development.

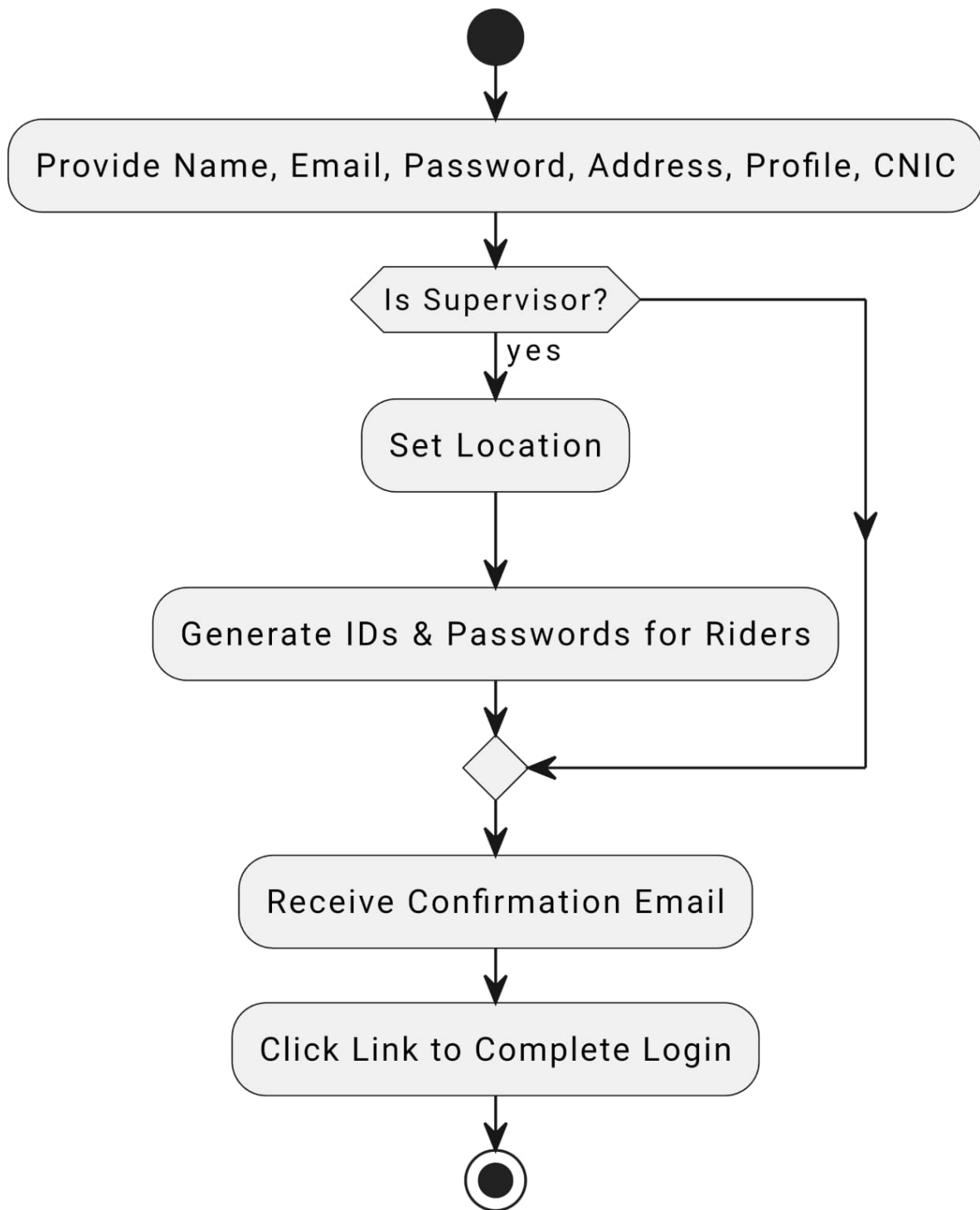
4. Proposed Model

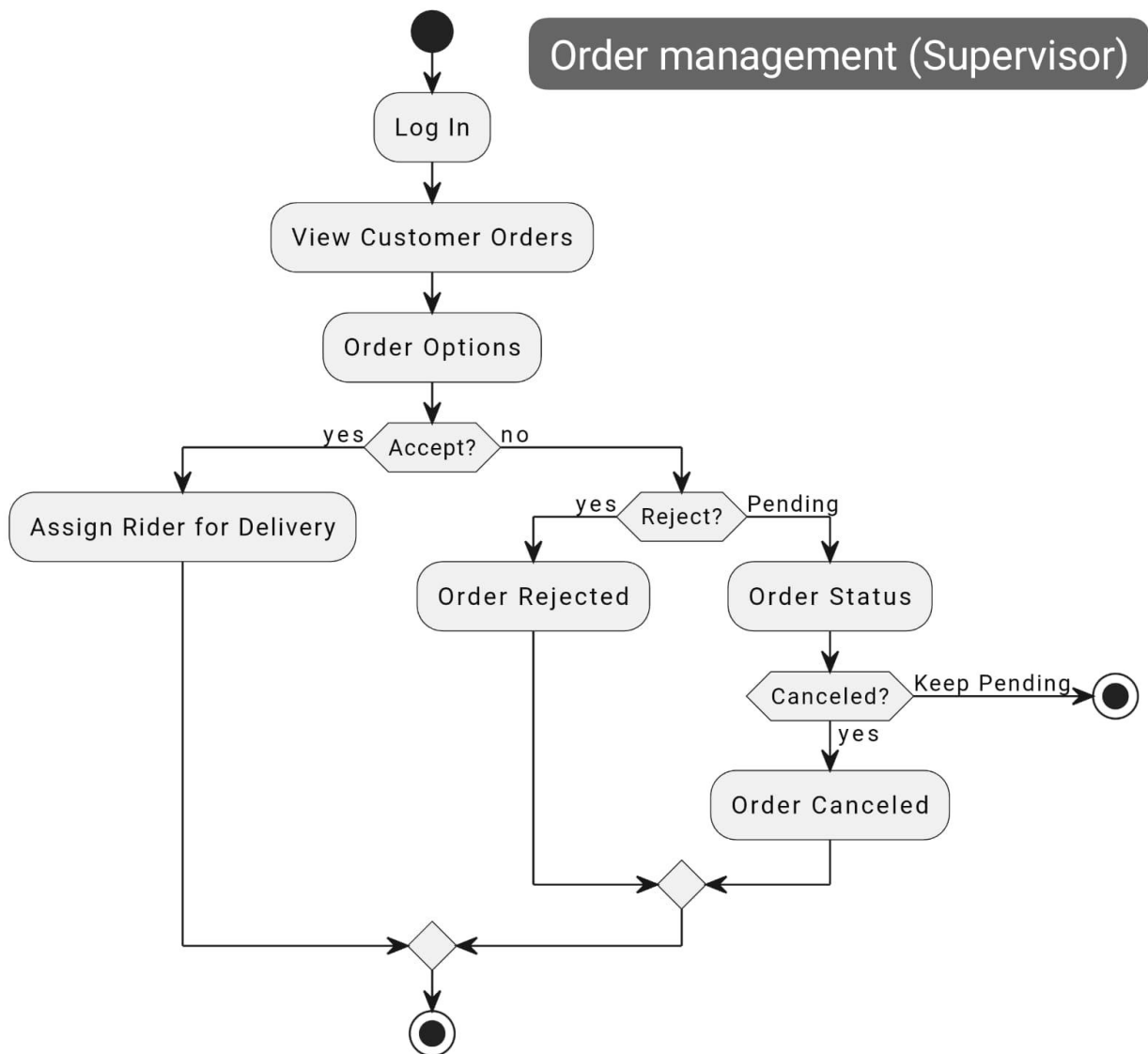
Project Flow Diagram











New Features

Types of Roles (Users)

1. **Customers:** Browse products, place orders, give feedback.
2. **Supervisors:** Manage inventory, process orders, register riders.
3. **Riders:** Manages Order Assigned by Supervisor

Functional Requirements (Detailed, Role-wise)

Customers

- **Registration/Login:** Secure authentication using Firebase.
- **Ro Plant Browsing:** View and search for Their Suitable Ro plant for assigning orders.
- **Order Placement:** Add products to cart and place orders.
- **Feedback:** Provide feedback on products and services.

Supervisors

- **Workflow:** seeing all order through his tab and selected the most feasible order where his rider can deliver with a short period of time.
- **Order Processing:** View and manage customer orders.
- **Rider Registration:** Register and manage delivery riders.
- **Feedback:** Provide feedback on products and services

Non-Functional Requirements

- **Scalability:** Ability to handle growing user base and data.
- **Performance:** Fast response times and smooth user experience.
- **Security:** Protect user data and ensure secure transactions.
- **Reliability:** High availability and minimal downtime.

Stakeholders

- **End Users (Customers and Supervisors)**
- **Developers**
- **Business Owners**
- **Investors**
- **Delivery Riders**

Hardware Specifications

For development and testing:

- **Development Machines:** High-performance computers with at least 16GB RAM, SSD, and modern processors.
- **Testing Devices:** Range of smartphones Android, tablets, and desktops to ensure cross-platform compatibility.

Chapter 4 Project Plan & Initial Design

Introduction

This chapter outlines the project plan and initial design for **Hydro Mart**. It describes the project planning process, breaks down the project into manageable modules with time estimates, and includes a Gantt chart to visualize the project timeline.

1. Project Planning

Description of How the Project Will Be Completed Within the Allotted Time

We will use Agile methodology same iterative development regular feedback and flexibility in Hydro Mart. By this approach, the management of the project team is transformed into adjusting to changes and developing a system further based on feedback from stakeholders. The project will be divided into many sprints for every sprint focusing on some of the modules and features needed in their application. To ensure that the project goes in right direction scrum meetings will be held with certain intervals where we can track what has been done and resolve if there are any blockers involved. Part of what Agile serves to do is, through its iterative process, build such a product while it changing-on-the-fly that the (somewhat static) application at the end can meet users' needs and expectations just fine.

Key Phases of the Project

1. **Requirement Gathering and Analysis:** The project requirements are established and the demands of the users are comprehended during this phase. To obtain user opinions and insights, surveys, interviews, and market research are conducted. Using the data gathered, comprehensive requirements that direct the design and development stages will be created
2. **Design:** In this phase, wireframes, UI/UX designs, and overall system architecture will be developed. The Design phase ensures that the application will at least be user-friendly, attractive, and technically sound. The wireframes will provide a basic layout regarding the application, while the UI/UX designs are geared toward enhancing user experience.

3. **Development:** The application will be designed and implemented in phases, and the number of modules that should be developed in each phase will be the basis of the phases. Before its official launch, it would contain basic functionality such as user identification and the ability to browse through products. Advanced tools and features of the current generation and designing methodologies will be incorporated to create a robust and fault-tolerant application.
4. **Testing:** After the development process, there will be some testing phases in order to notice whether there are some imperfections, glitches, or errors in the application. The testing phase will also consist of the unit testing, integration testing and the user acceptance testing.
5. **Deployment:** After the application has been developed and the functionalities tested and optimized the application will run on Android platforms. The deployment phase consists on installing IT structures, defining the servers to be used and making the application ready for the users.
6. **Maintenance and Updates:** Subsequent to the application's launch, there will be constant assessment of its efficiency, ironing out of glitches, and propelling of new versions. The maintenance phase works to guarantee that the application is fully operational and relevant to users' requirements and other advancements in technology.

Project Milestones

- **Milestone 1:** Completion of requirement gathering and analysis.
- **Milestone 2:** Finalization of design and architecture.
- **Milestone 3:** Development of core features (authentication, product browsing).
- **Milestone 4:** Integration of Tsp algorithm.
- **Milestone 5:** Completion of testing and bug fixing.
- **Milestone 6:** Deployment and launch of the application.

2. Module Breakdown and Time Estimates

Analysis and Time Estimates for Each Module

Here are the primary components of the project, the tasks to be completed for each, and estimated time: Scheduling also benefits from this breakdown because it helps in proper allocation of resources so as to keep the project on track. This helps in making a proper plan for tackling them because they are identified right from the start so that they do not take a lot from you.

1. Requirement Gathering and Analysis

- **Tasks:** Carrying out surveys, collecting users' feedback and defining requirements.
- **Time Estimate:** 2 weeks

2. Design

- **Tasks:** Create wireframes, design the UI/UX, and develop the overall system architecture.
- **Time Estimate:** 3 weeks

Development

a. Authentication

- **Tasks:** Setup user registration, login and authentication with Firebase. Make certain that suitable and solid levels of authentication are in place.
- **Time Estimate:** 2 weeks

b. Ro Plant Browsing and Order:

- **Tasks:** Enable customers to browse nearby RO plants and place orders using Google Maps integration. Focus on ease of use and accuracy in locating the nearest RO plant.
- **Time Estimate:** 3 weeks

c. Order Management

- **Tasks:** Design features of order placement and its management to track and control all the orders within the assigned riders. Develop functionalities related to order management, its status, and the rider in charge.
- **Time Estimate:** 3 weeks

Testing

- **Tasks:** There is a common procedure that should be followed: unit tests, integration tests, and user acceptance tests should be performed to ensure the application's free of bugs and meets the end-users' expectations.
- **Time Estimate:** 3 weeks

Maintenance and Updates

- **Tasks:** Take measures to check the application's performance and effect the required corrections to remove bugs and release updates frequently to ensure a perfect and updated application.
- **Time Estimate:** Ongoing

Identification of Challenging Modules

- **Tsp Algorithm Integration:** The merging of the Traveling Salesman Problem (TSP) algorithm is distinguished as being a sensitive module since it must guarantee correct and optimal delivery routes. The TSP algorithm is a special module as it defines delivery paths and contributes to the minimization of delivery time, thus being an important part of the order management system.
- **Real-Time Data Management:** The main struggle is to keep the data sync fast and work in real time across multiple devices while working with a large number of orders and users. This is especially important for maintaining the application's performance since all the users must see the same information in real-time.

Gantt Chart

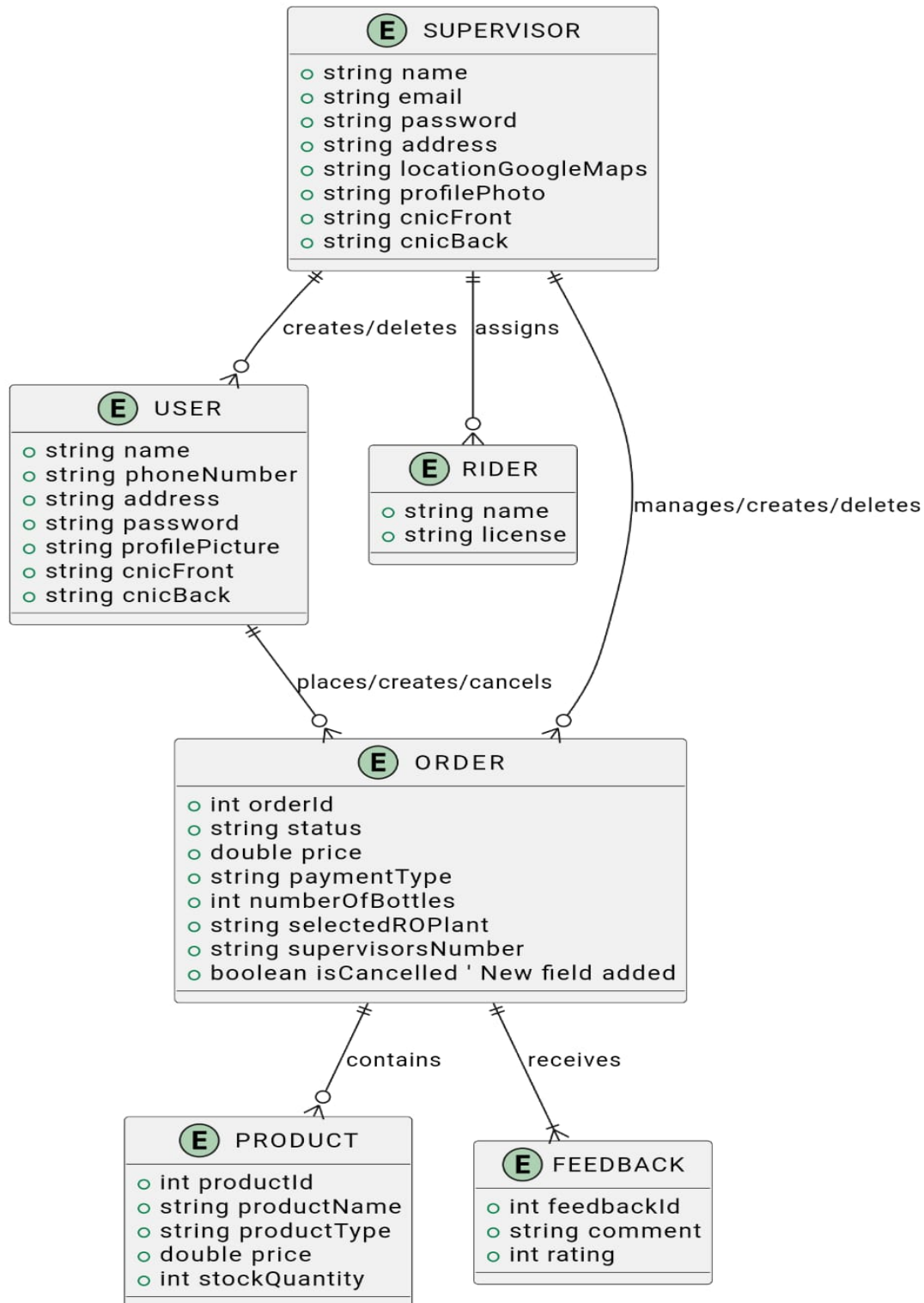
HYDRO MART

Gantt Chart

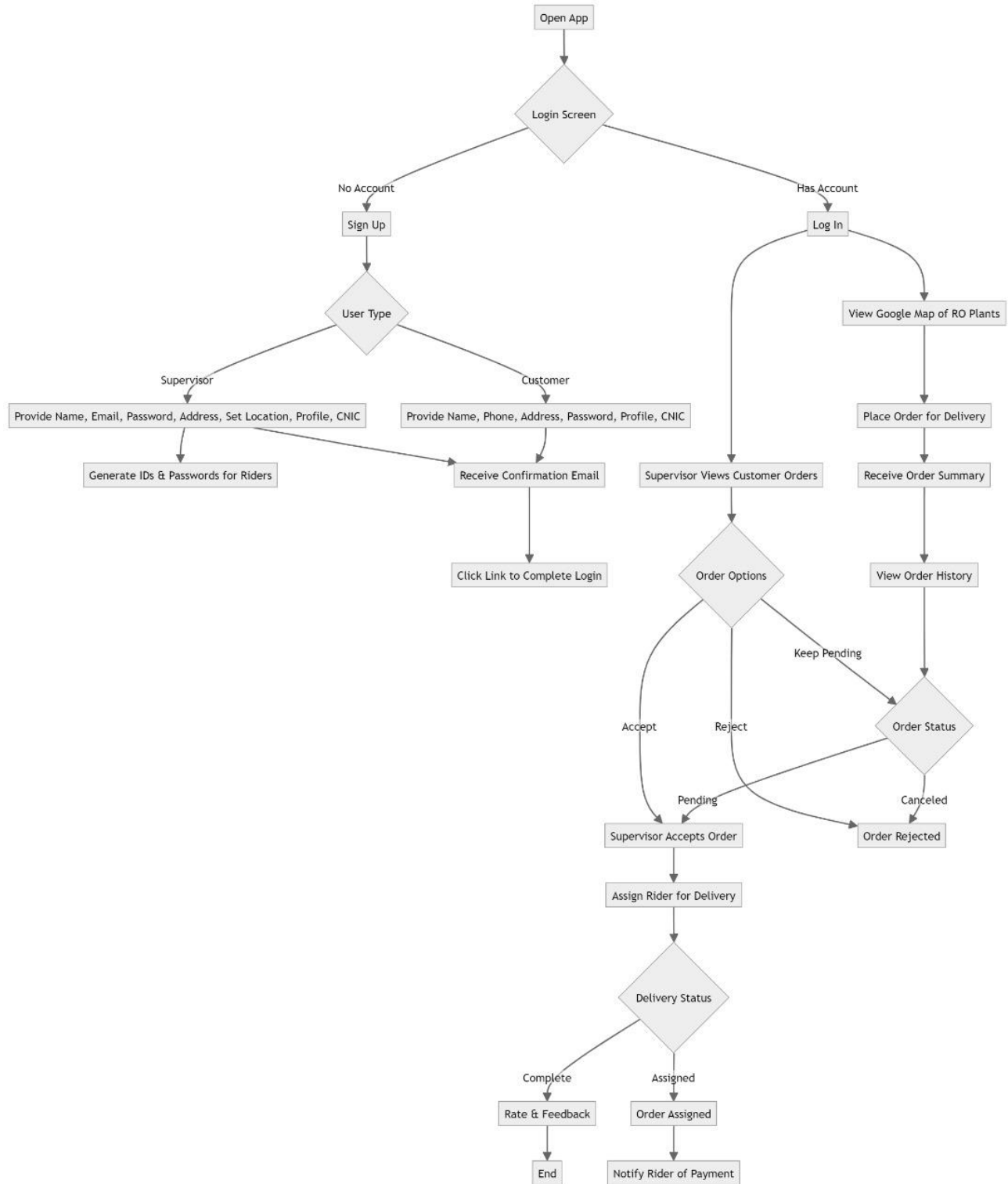
PROCESS	QUARTER 1						QUARTER 2					
	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Project Planning												
Requirements Gathering												
Research On Project												
Front End Development												
Back End development												
Deployment												

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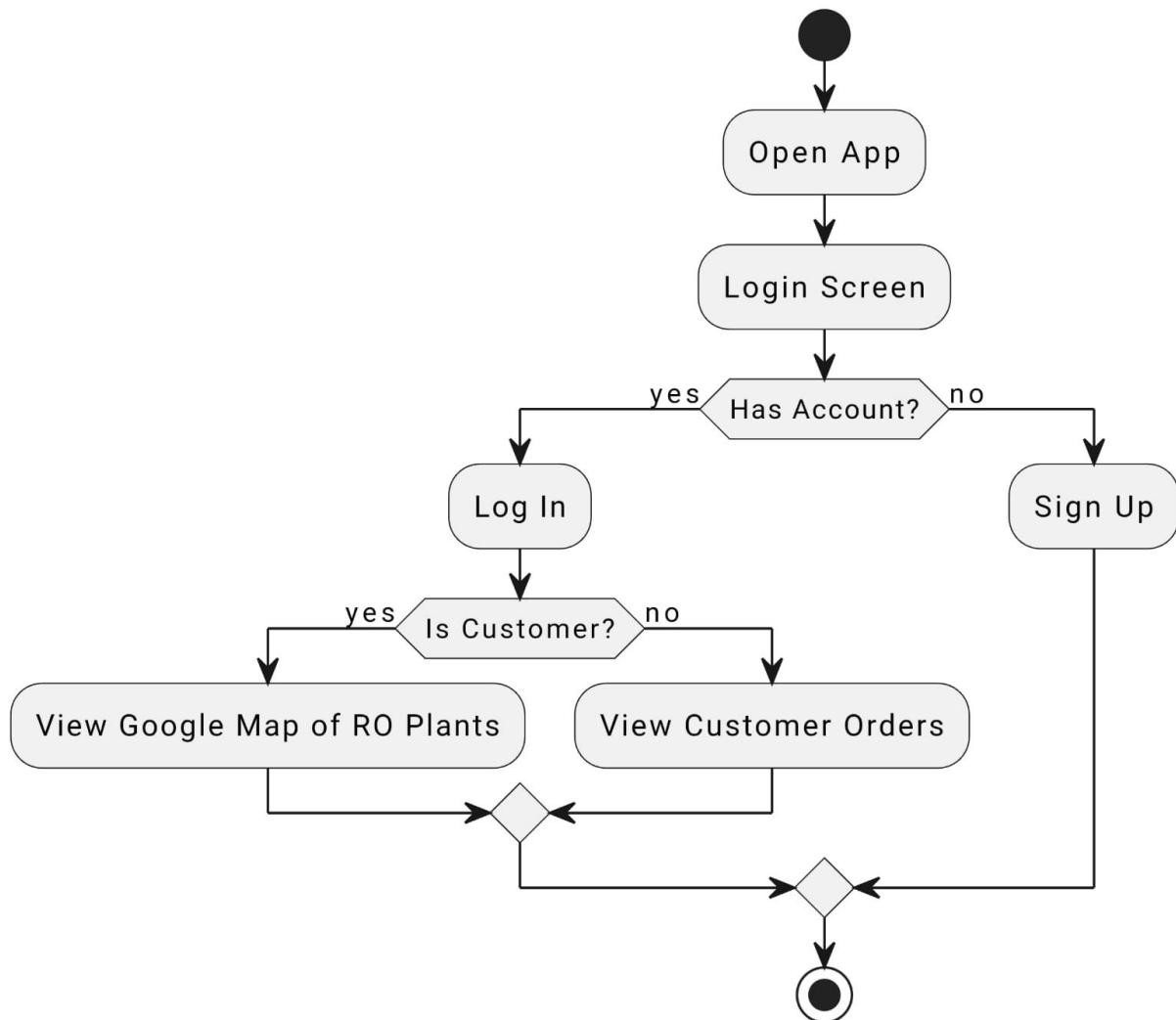
ERD DIAGRAM :

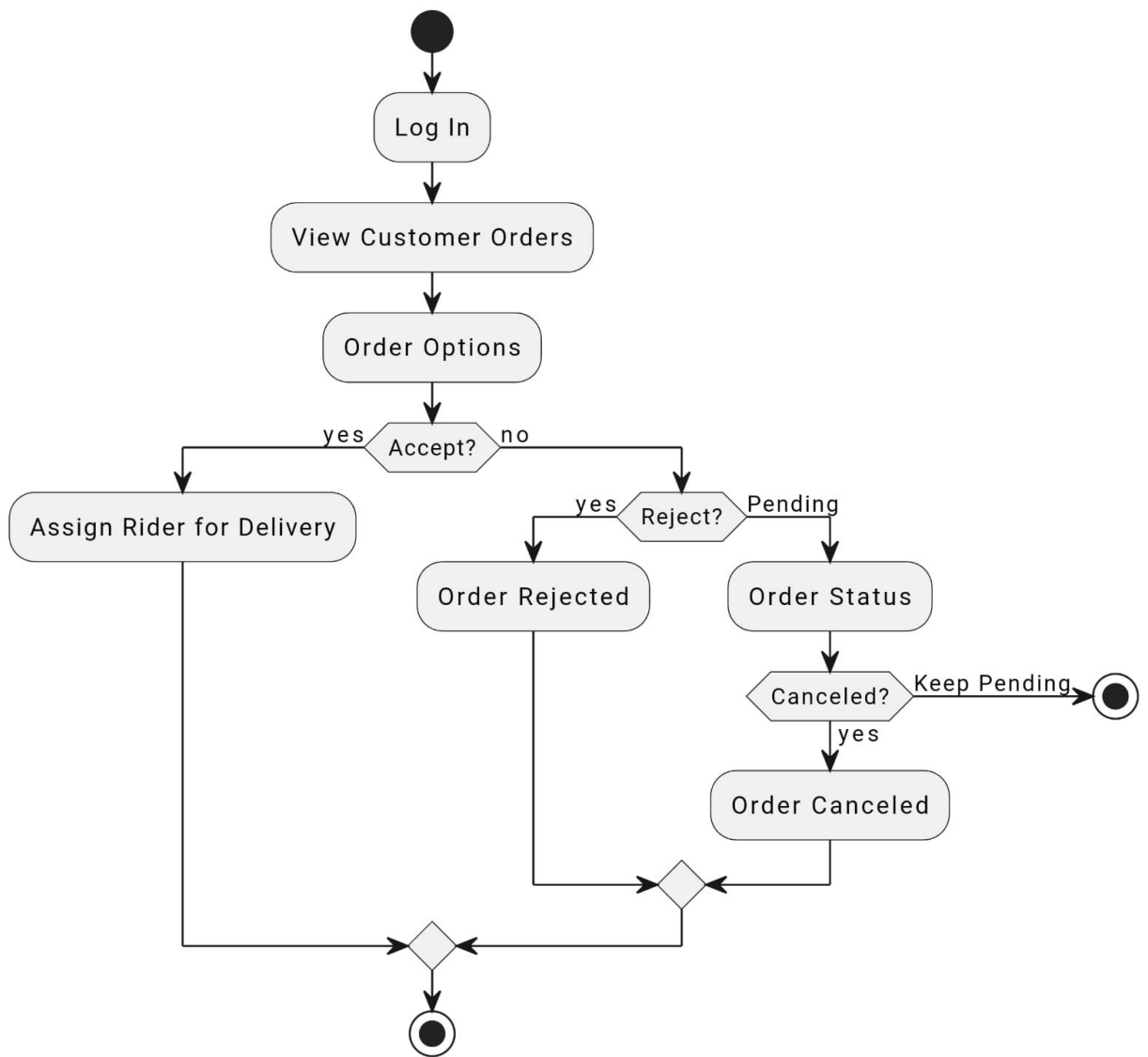


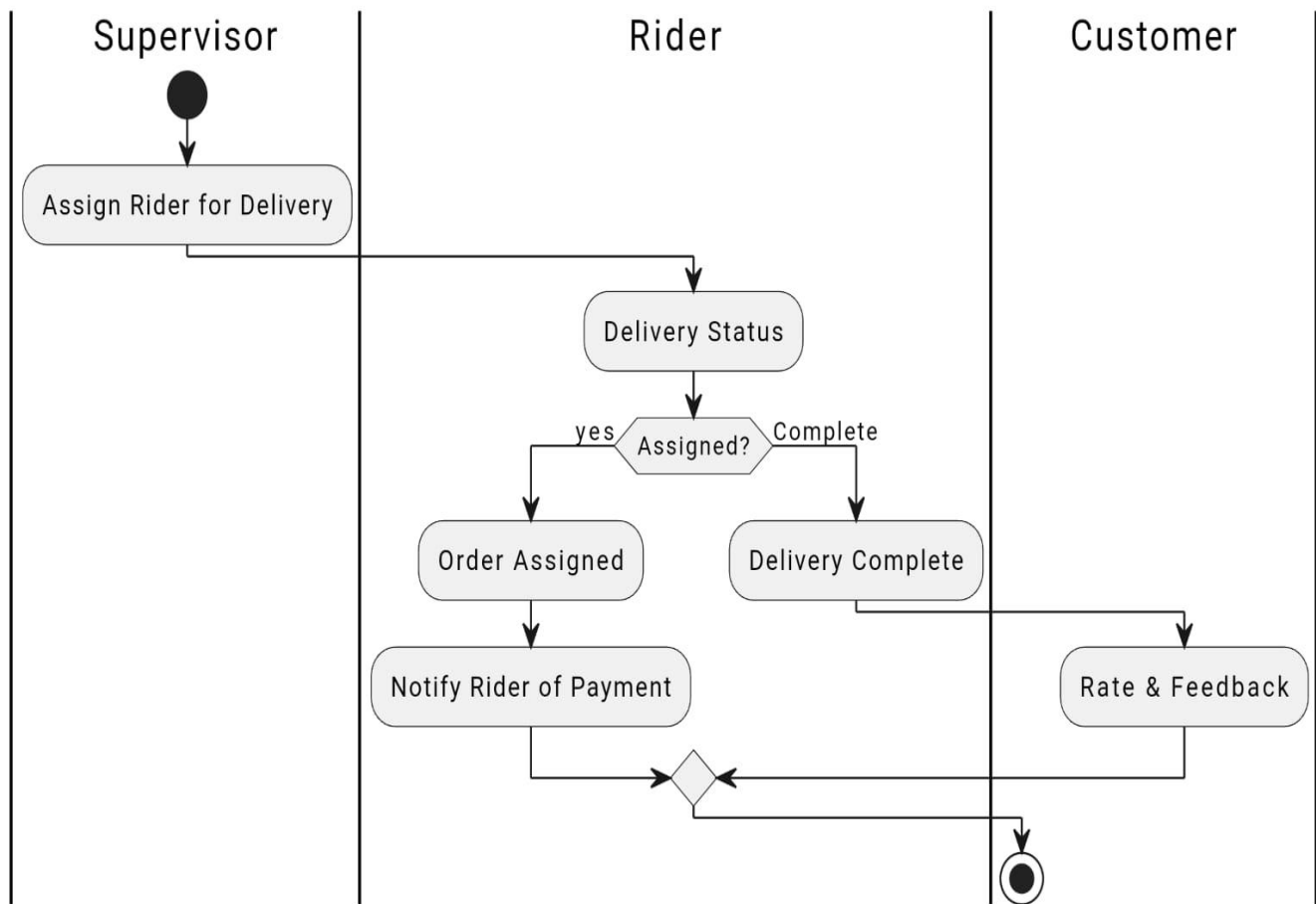
FLOW DIAGRAM:

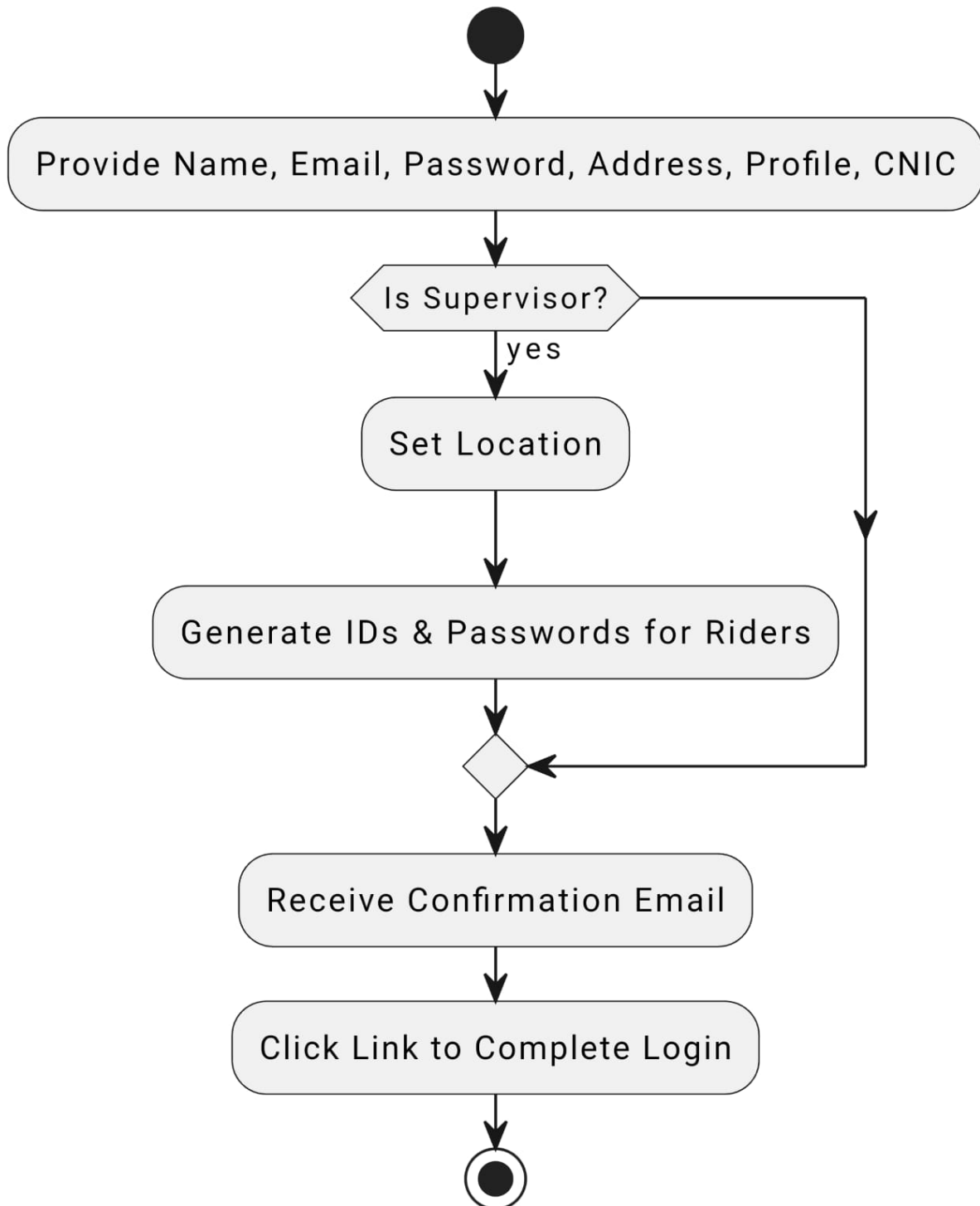


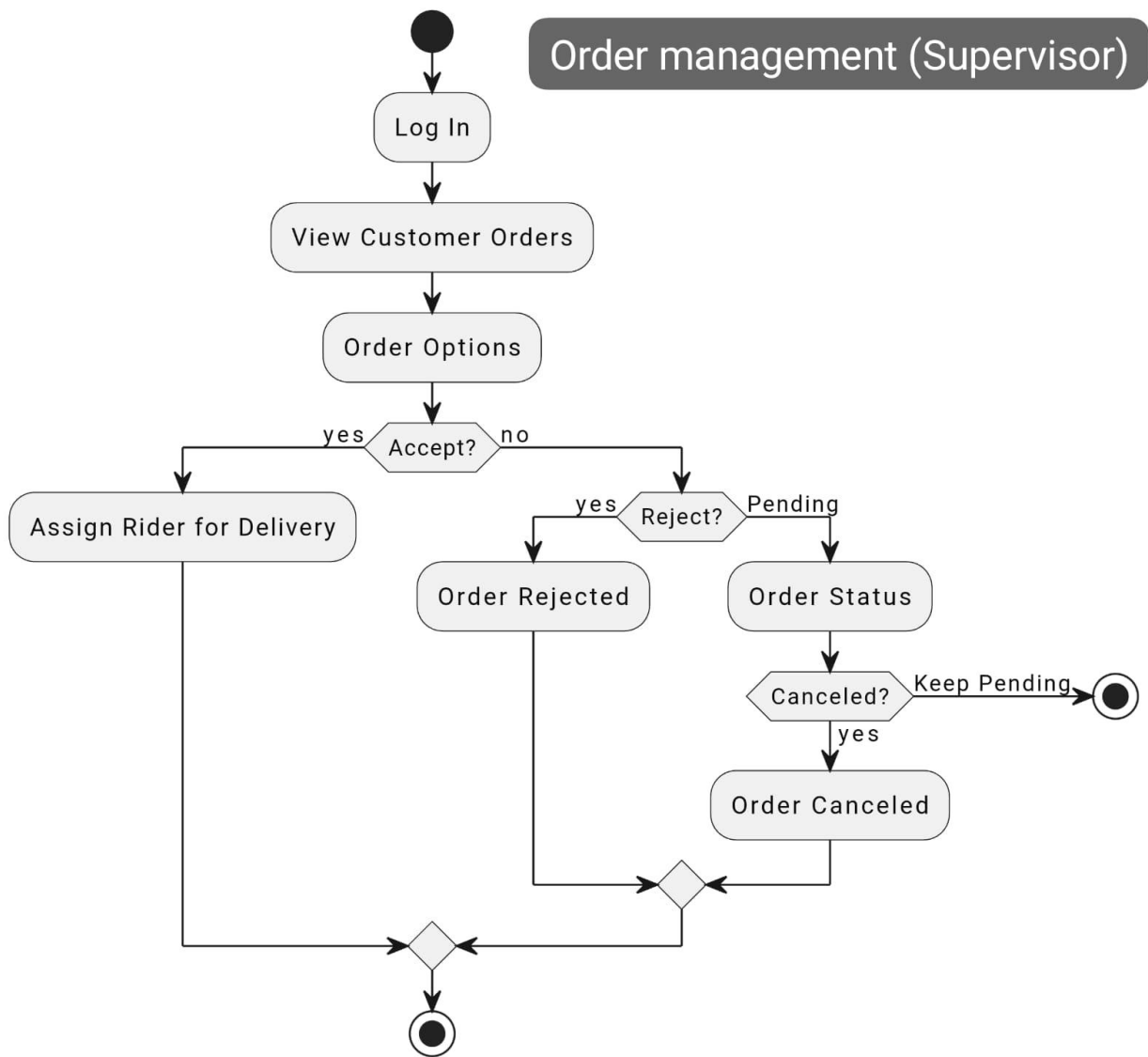
ACTIVITY DIAGRAM:











Chapter 5: Project Design & Development

Design

Front End Design

Color Theme:

- The application will have the same set of colors across all platforms. An eye-catching display but user-friendly. Primary colors are [insert primary colors, for instance, Blue and White], chosen in its nature for readability purposes.

Logo:

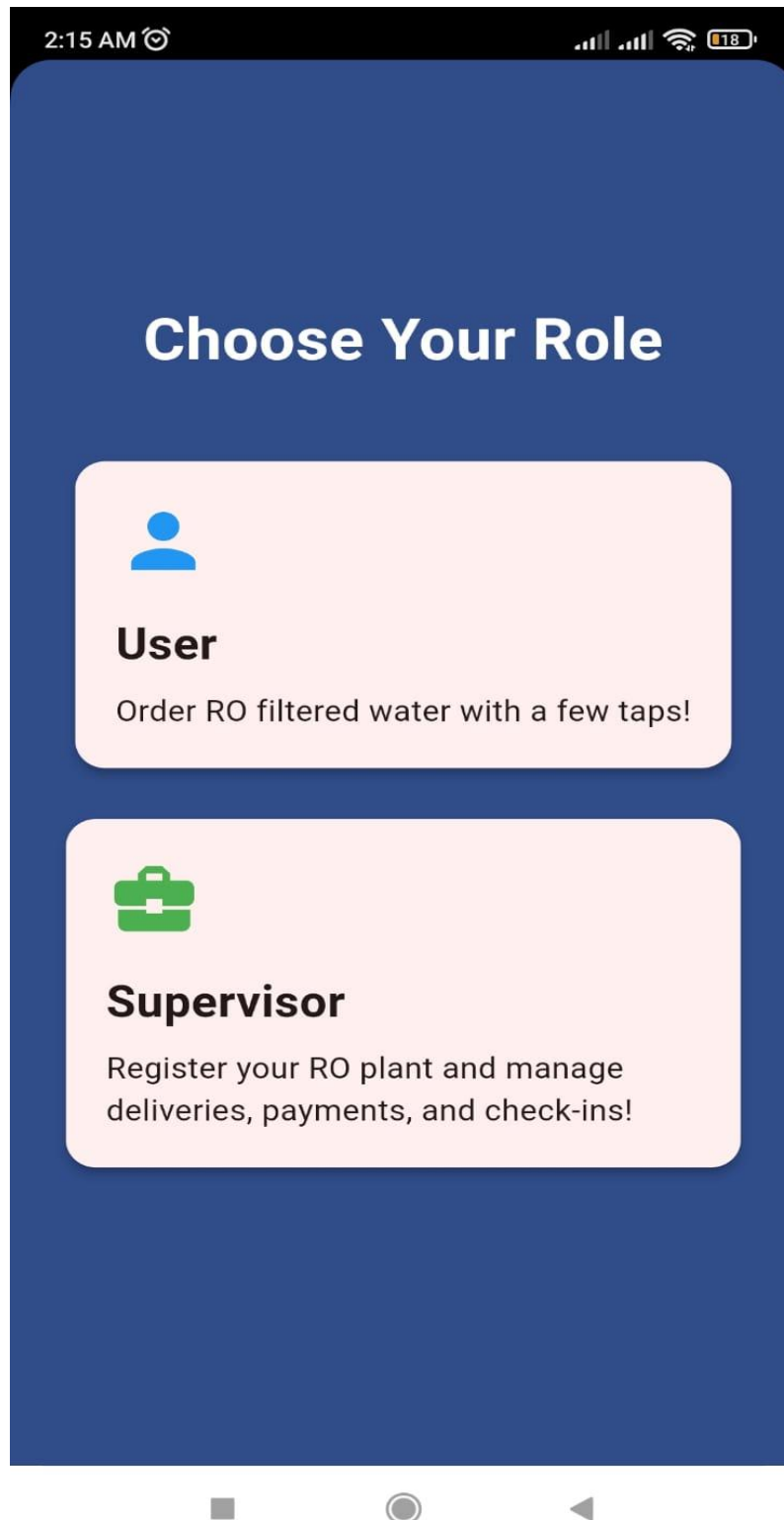
- This is a simple design that speaks to the brand identity of Hydro Mart. There's a droplet of water, which symbolizes purity and refreshment.



Template:

- Design pattern of the UI template: a modern, clean design pattern infused with usability. The template includes elements such as navigation bars, product cards, and buttons, styled with the chosen color theme.

- Screenshots of the UI:



2:08 AM 🔒 📶 🔋 22

Login Screen

Email

Password

Login

Not Registered Yet?

[Go to Register Screen](#)

Login as a Rider

2:08 AM   

← Register as a Supervisor

Email

Password

Address

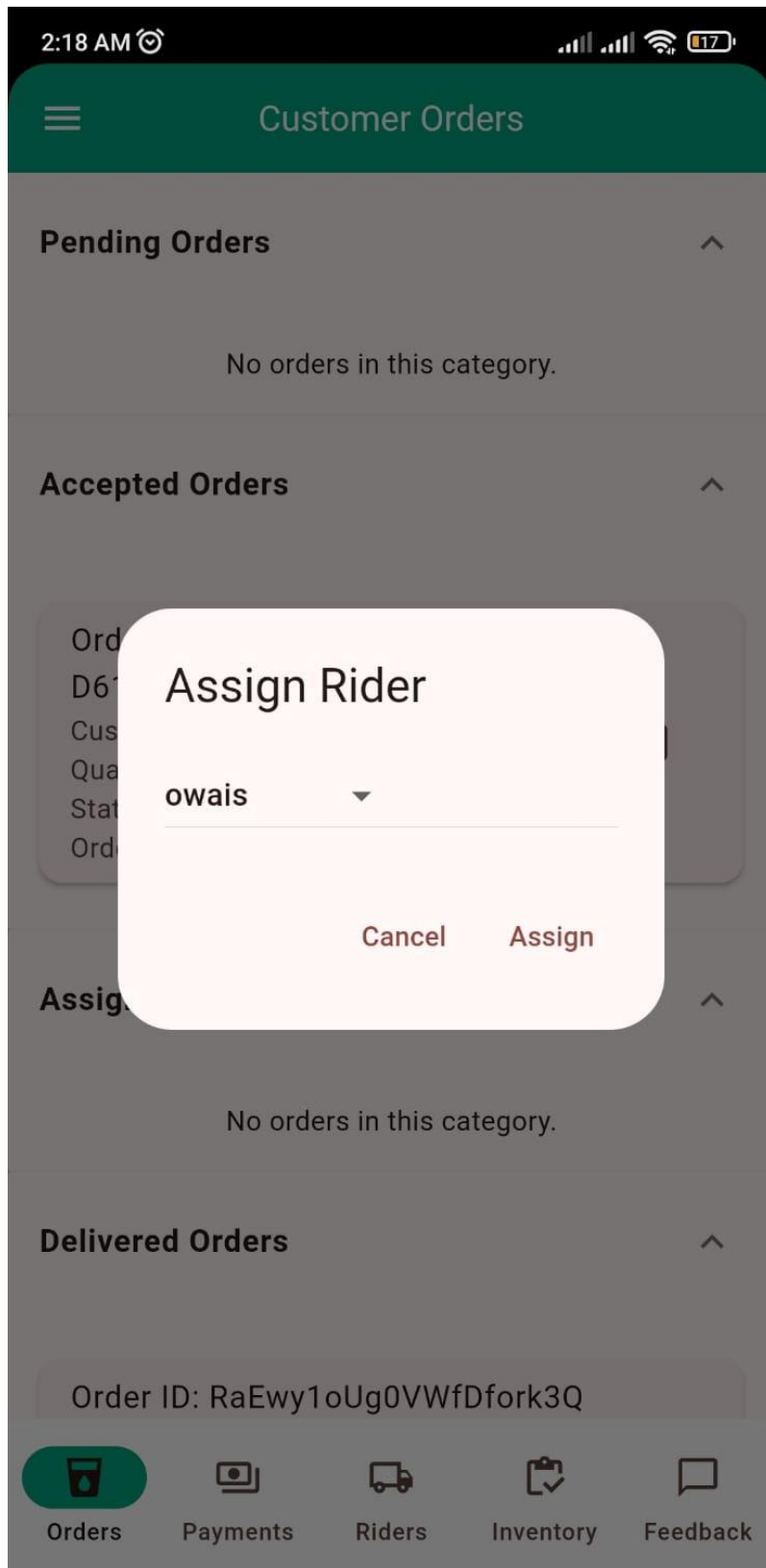
▼  +92 Phone Number

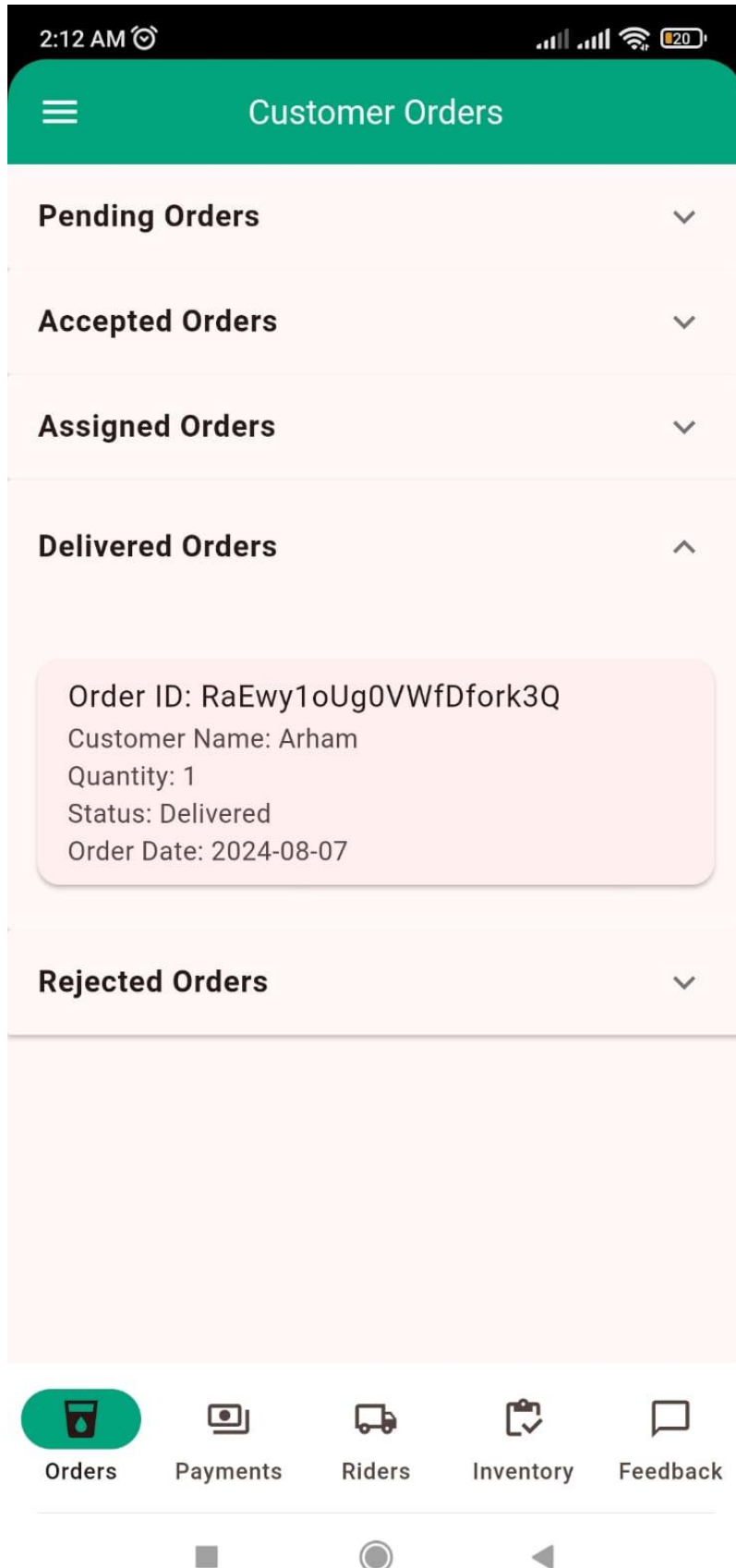
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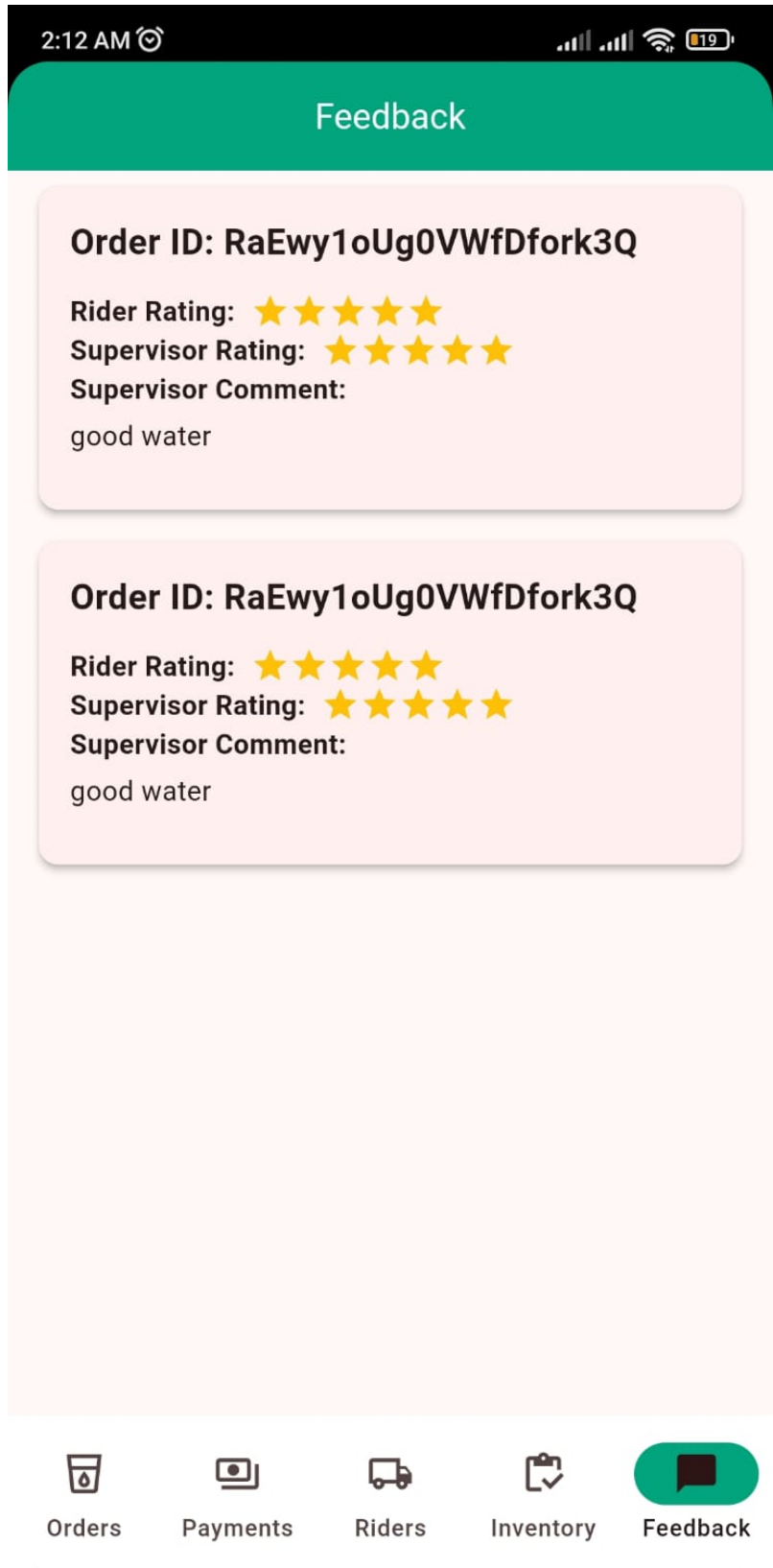
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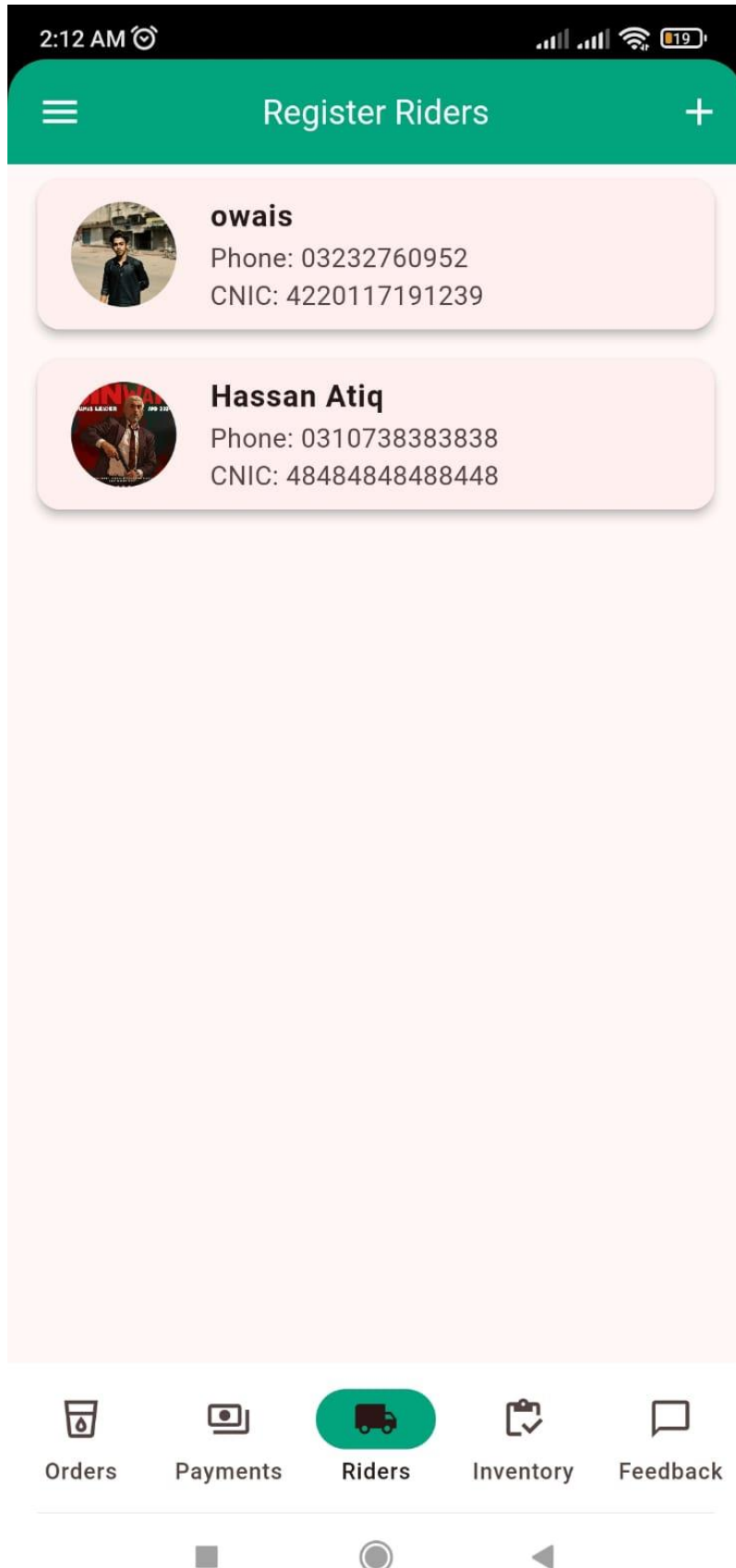
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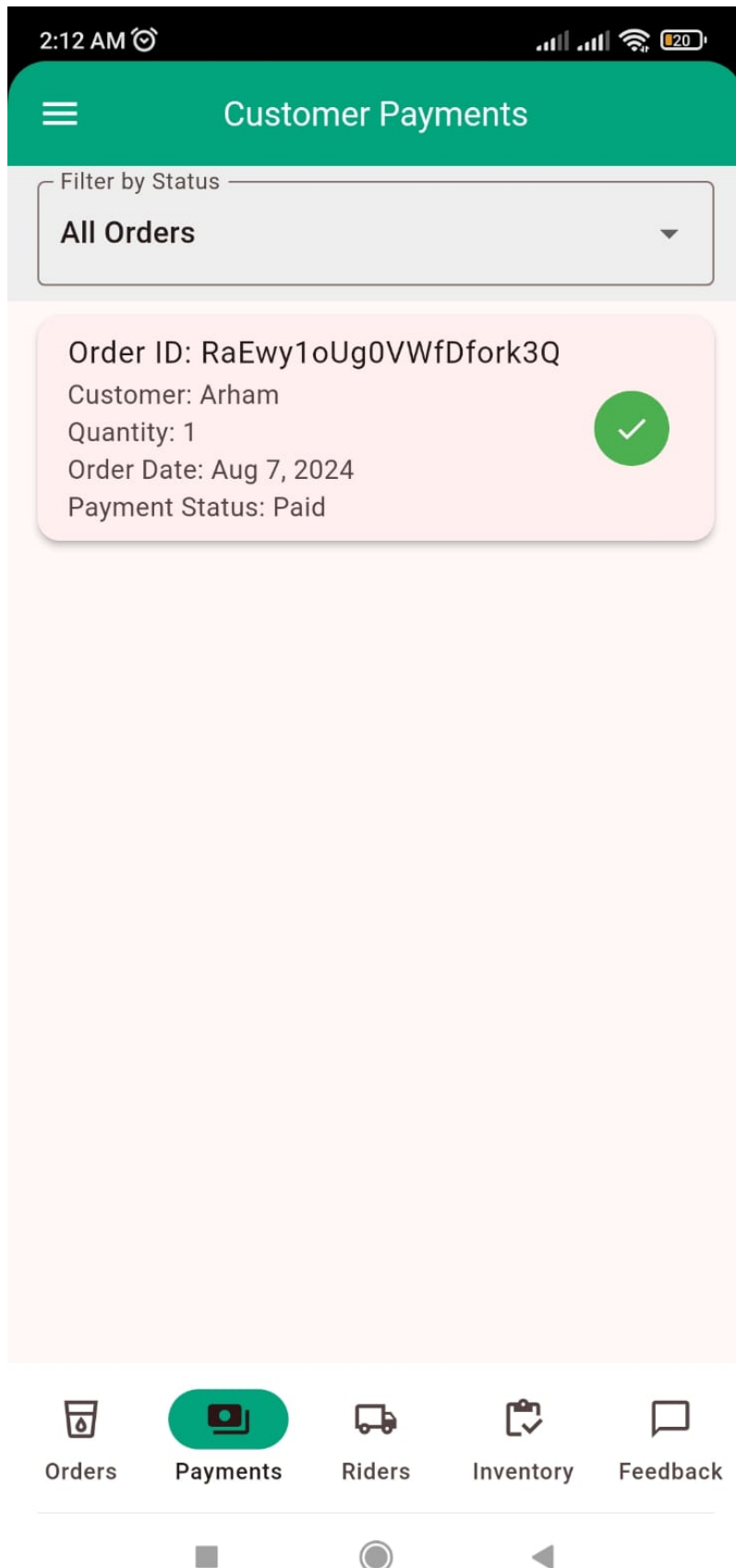
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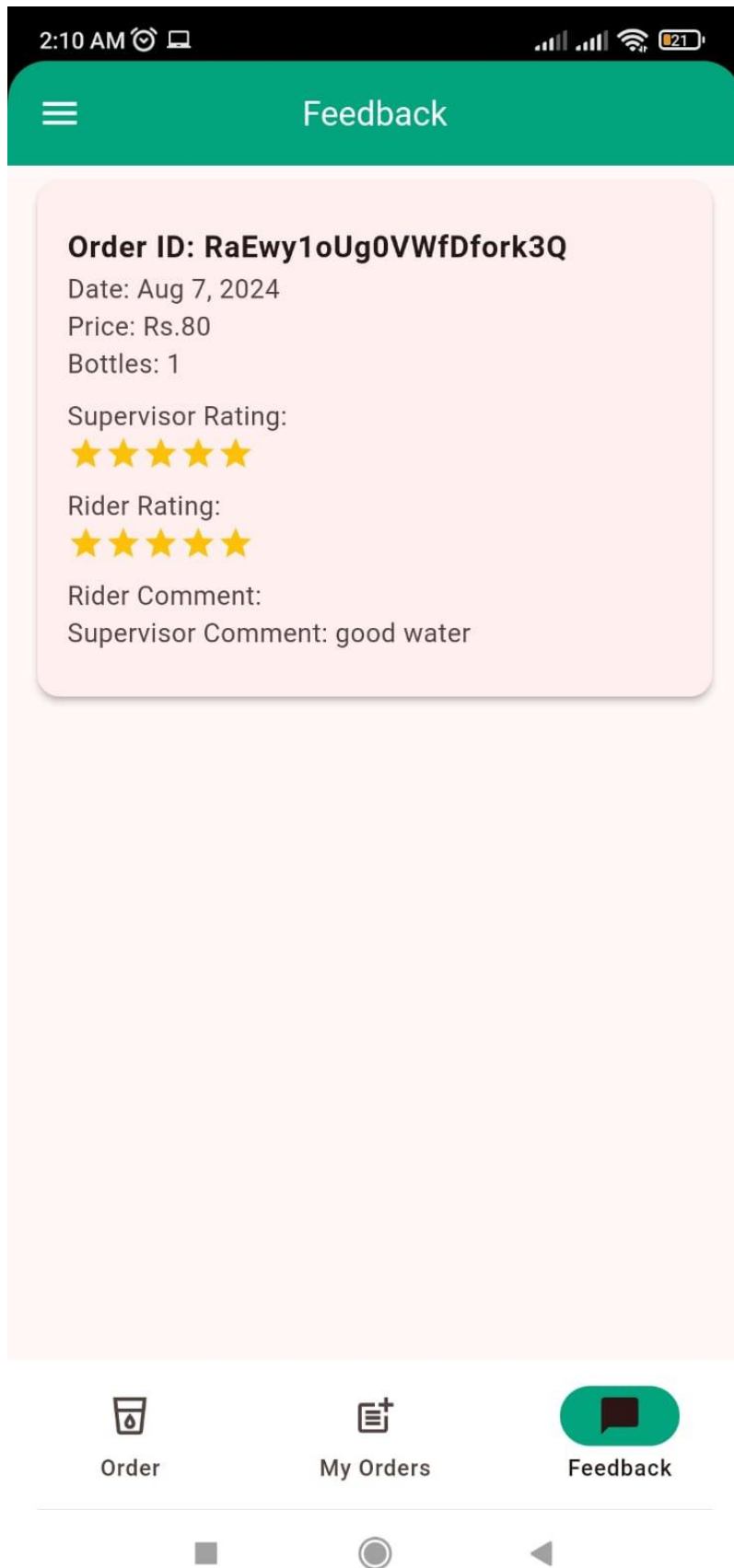


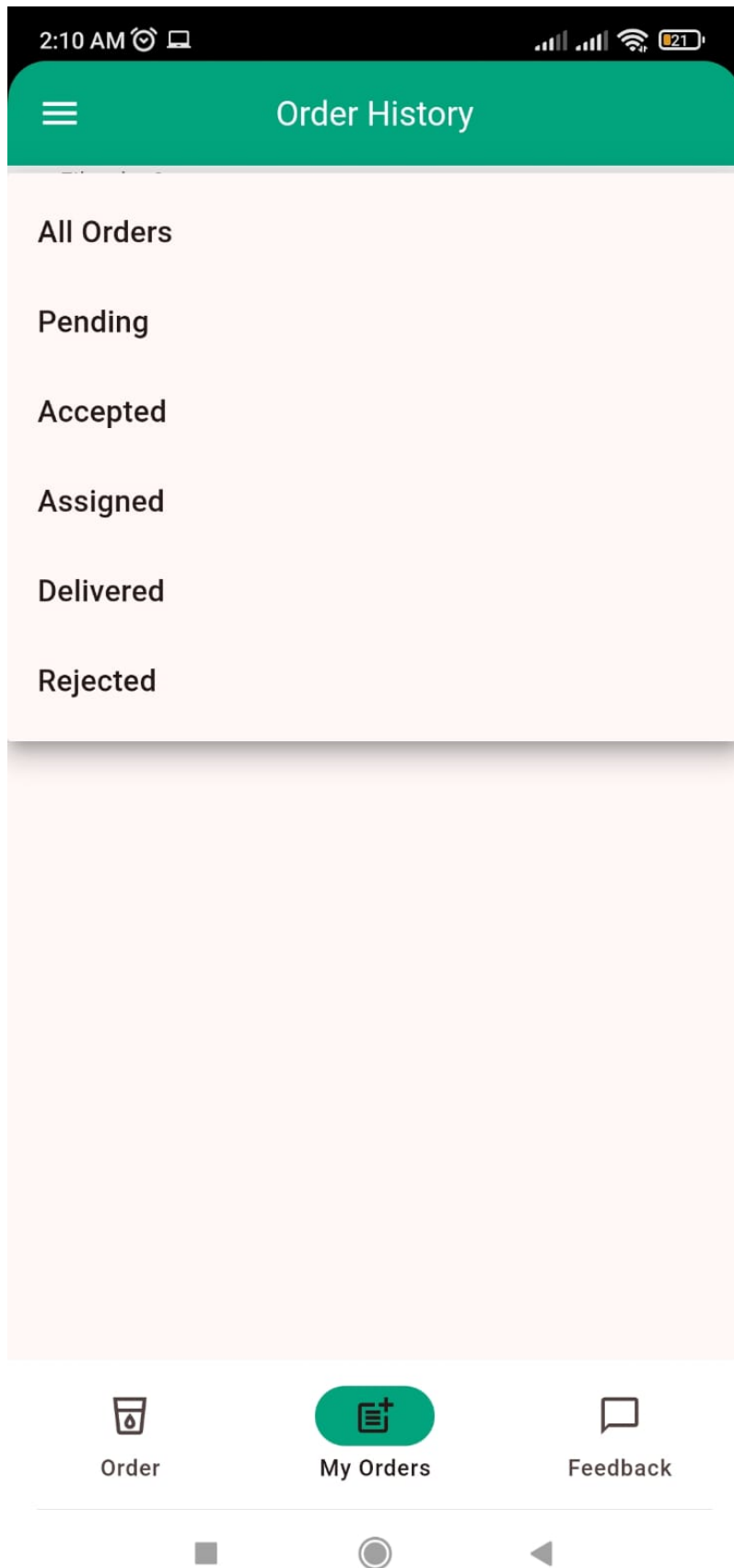


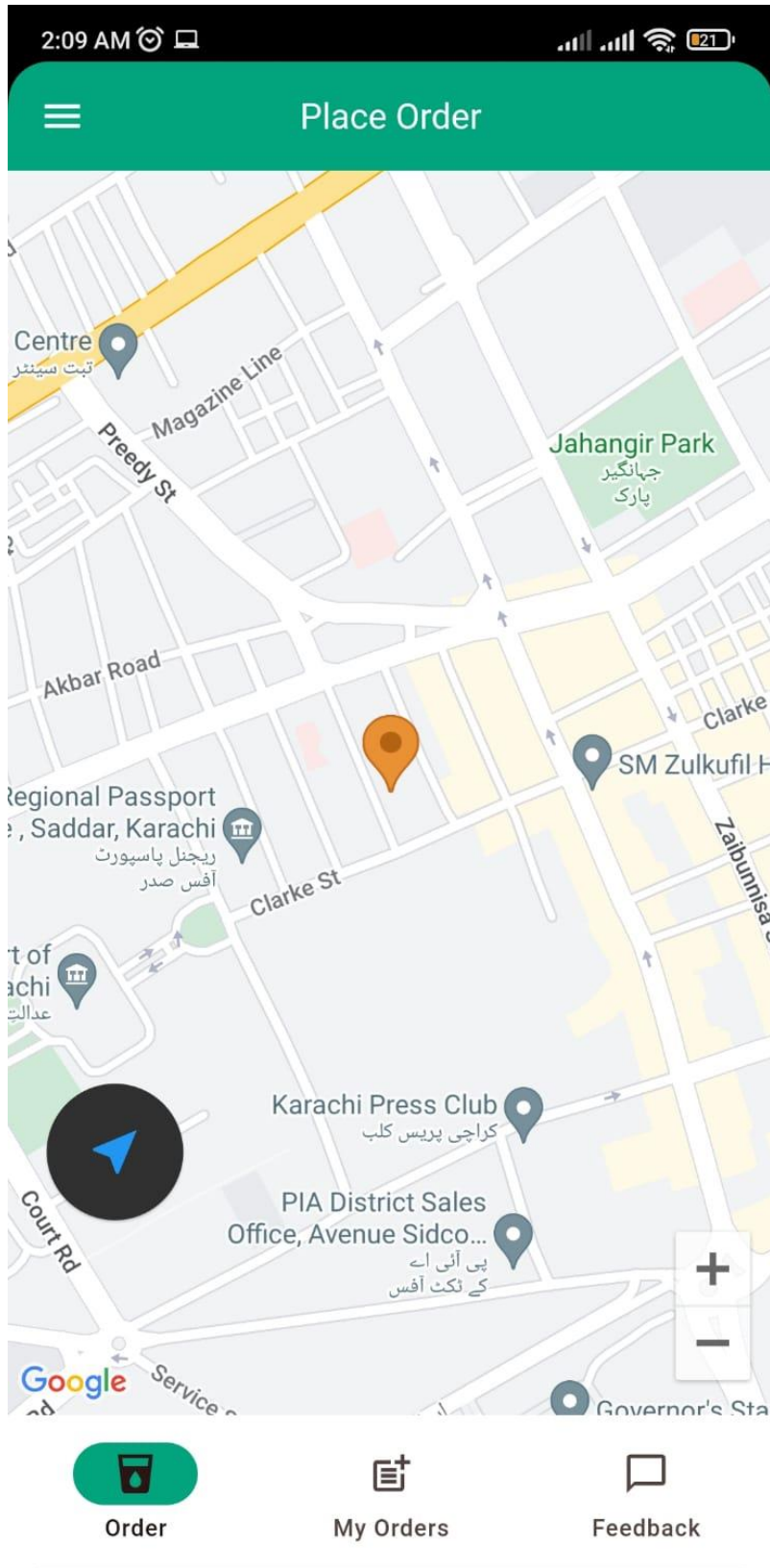






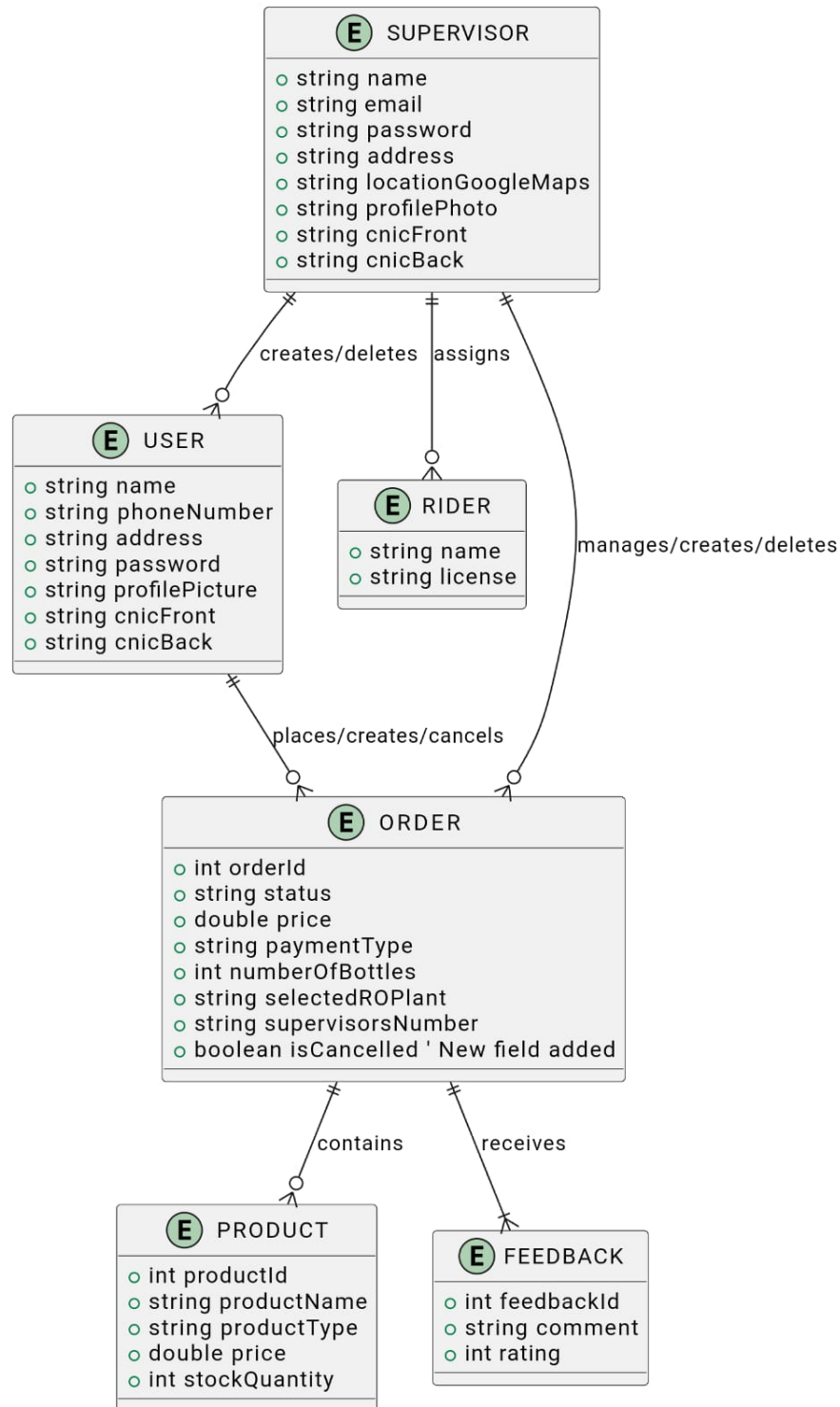






Backend Design

\Physical ERD of the Database:



Development

Development Stages

Stage 1: Setting Up the Environment

- The development began with setting up the Flutter environment, including IDE configuration and dependency management via pubspec.yaml.

Stage 2: UI Implementation

- The front-end components were developed using Dart and Flutter widgets. The MaterialApp and Scaffold widgets form the backbone of the UI, with custom widgets for product cards and navigation.

-Stage 3: Backend Development

- The backend logic was implemented using Firebase as a backend-as-a-service (for authentication and database management. The app integrates Firebase Authentication for secure user login and registration.

-Stage 4: Integration

- Integration of the UI with backend services, including real-time data syncing and state management using Provider

Libraries Used

Flutter: Version 3.0.5

Dart: Version 2.17.1

Firebase Core: Version 1.10.6

-Provider: Version 6.0.2

-Backend Development:

In the implementation of the backend logic, Firebase was used as a backend-as-a-service. Most of its features were targeting authentication and database management. Firebase Authentication provides safe user login and registration in the app.

Chapter 6: Test Cases

6.1 Introduction to Testing

6.1.1. Ensuring Functionality

- **Requirement Verification:** Testing involves checking if the developed software has met the functional requirements specified by the stakeholders. This confirms that each of the components and capabilities performs the activity it was made to do.
- **User experience:** This would help the developers in confirming that it is, in fact an easy-to-use application and can perform optimally during various situations whenever the application is used like how the end users would use it.
- **Compatibility:** One can test in order to be very sure that all the software is intact; it will perform to the expected standards across devices, operating systems, and browsers to support the users.

6.1.2. Enhancing Reliability

Error Detection: Every testing type, including unit, integration or system testing, allows the developers to screen out problems before it is seen by end users. The process minimizes the factors that contribute to crashes and failures when using the application of software in real life situations.

Consistency: Testing is beneficial to ensure stability of the developed software since these new changes do not propagate other new problems. Regression testing is one of them especially it mandates that: should be equally credible as the software under test grows.

Performance: Performance testing can be deemed as a stress testing that are supposed to check whether the software in question is going to perform at its best depending on the amount of working load it has been subjected to without any slowness.

6. 1. 3. Ensuring Security

•**Vulnerability Identification:** Security testing facilitates identification of the gaps, vulnerability and risks for instance security infringements and Company information loss before the software is released. This enhances user satisfaction as their data is safe from accessing by other people who are not authorized.

Compliance: Presently, numerous firms have set policies that ought to be adhered to with regards to the management of data and information. It also aids in establishing that the developed software complies with the statutes, hence does not invite adverse legal/ financial repercussions.

Resilience: Security testing also assesses how the software's defence measures and methods by which it can protect itself when facing such a threat, followed by the methods through which the software can rebuild itself if threatened.

6.1. 4. Before deploying the application some problems could be pointed out and corrected.

- **Cost Efficiency:** Problems that can be solved at the early stages involving development are cheaper to solve than when the programs are fully deployed. Testing is carried out right from the initial stages of the project because the costs that are incurred are relatively low, and it also saves time.
- **Reduced Risk:** Carry out of tests assist reduce the risks of developing serious issues with the actual production of the product for instance the costs, reputation and consumer complaints may be significantly impacted.
- **Continuous Improvement:** Testing helps in assessing the implementation of the software as well as identifying vulnerability to the stakeholders thereby helping in continuous improvement of the software. These performances of the parts lead to a higher quality of the final product due to this process of testing and rehearsing.

6.2 Testing Strategies and Methodologies*

Unit Testing:

The kind of tests that were created included those targeting a specific area of the application that could be a function, class, or method. These tests helped in that they verify that when these components are separated out, they function correctly.

Example: For the task such as adding products into a cart, a unit test was developed merely to test if the correct product of the right quantity was stored.

Integration Testing:

The tests which involved combining the various sub-modules and components are considered integration tests. This was very important so that all sub-modules like authentication and management of products could work seamlessly.

Example: To ensure that interdependency between the Flutter UI and the Firebase authentication service was tested, integration tests were conducted.

System Testing:

In system testing, the entire application was tested since the test was on the integrated and complete application system. The aim was to guarantee the adequate compliance of the integrated system with all the requirements and the correct behavior in a context.

Example: Integration tests focused on the proper functioning of each software component and system tests were carried out with the intention of emulating the use of the site to browse products, to place an order and to make a payment.

Acceptance Testing:

After integration testing, there was only acceptance testing in which the application was checked against the business requirements to be implemented. This phase made sure that a given soft ware was properly developed and ready for release.

Example: Stakeholders were used to carry out the testing of the application to ascertain whether the different features like the product search and the check out button were satisfactory.

6.3 Test Environment Setup

Hardware Specifications:

Testing was carried out on devices with varying specifications, including low-end Android phone to ensure compatibility across all target platforms.

Software and Tools:

Development Tools: Flutter SDK, Android Studio were used for developing and running the application.

Testing Tools: Firebase Test for and Android Emulator were employed for testing on different platforms.

Network Configurations:

- Tests were conducted under different network conditions, including high-speed Wi-Fi, mobile data, and limited connectivity, to ensure the application's performance under various scenarios.

Specific Conditions:

- The application was tested in environments simulating high user traffic and low memory availability to assess its performance and stability.

6.4 Test Case Design

SN O.		ACTUAL O/P	EXPECTED O/P	STATUS
1		Incomplete email	Incorrect format	pass
2		Unregistered email	Registered email	fail
3		Correct password	Login successfull	pass
4		correct password	Login failed	fail
5		Incorrect password	Login successfull	fail
6		Incorrect password	Login failed	pass
7		Open camera	Opens gallery	fail
8		Opens gallery	Open gallery	Pass
9		Create order	Not create order	Fail
10		Create order	Order created	Pass

6.5 Test Case Documentation

Detailed documentation was created for each test case:

Test Case Description: Verify that a user can successfully log in with valid credentials.

Preconditions: The user must have an existing account with valid credentials.

Test Steps:

1. Open the application and navigate to the login screen.
2. Enter a valid email and password.
3. Click the "Login" button.

Expected Result: The user is redirected to the home screen.

Actual Result: The user is redirected to the home screen.

Status: Passed

Comments: None

6.6 Test Execution

The test cases were executed in a planned sequence:

Sequence of Tests:

Unit tests were executed first to verify individual components, followed by integration tests to ensure module interaction. System and acceptance tests were conducted at the end to validate the entire application.

Timeline:

The testing process spanned over four weeks, with each type of test (unit, integration, system, acceptance) allocated specific time slots.

Challenges Encountered:

Some challenges included dealing with platform-specific issues, such as differences in how Android handle push notifications. These issues were resolved by implementing platform-specific code adjustments.

6.7 Test Results and Analysis

Analysis:

The majority of the test cases passed successfully, indicating that the application is stable and meets the specified requirements. The few failed cases were related to minor UI inconsistencies on different platforms, which were subsequently addressed.

Major Issues:

A significant bug was identified in the checkout process, where under specific conditions, the order was not being saved correctly. This was due to a race condition in the database query, which was fixed by implementing proper locking mechanisms.

6.8 Bug Reporting and Tracking

Tools Used:

Jira was used for bug tracking, providing a centralized platform for reporting, prioritizing, and resolving bugs.

6.11 Test Cases Conclusion

The testing process for the **Hydro Mart** application was thorough and systematic, ensuring that the final product was of high quality. The combination of unit, integration, system, and acceptance testing provided a comprehensive evaluation of the application. The testing methodologies used were effective in identifying and resolving issues, resulting in a stable and reliable application.

While some minor issues were encountered, they were addressed through careful testing and bug tracking. Future testing will focus on new features and continuous improvement of the existing functionalities.

Chapter 7: Conclusion & Future Work

Conclusion

Summary of the Project

The main objectives of the Hydro Mart project recapitulated with the ways in which they were achieved.

Outline the major features of the application, including efficient water delivery, order management, secure transactions and managing rider orders.

Discussion of challenges encountered during development and how they were overcome.

Impact of the Project

Describe how the Hydro Mart application can potentially affect the water delivery industry in urban areas. Explain benefits both for the user and for supervisors and delivery personnel, including increased efficiency, cost savings, and improved customer satisfaction.

Project Evaluation

Evaluate the overall success of the project in achieving its goals

Reflect on the feedback received from users and stakeholders in the course of testing.

Describe the limitations one faced while working on the project and how they were overcome.

Future Work

Possible Improvements

Point out the inadequacies of the present system with respect to additional features or enhanced existing features.

Bring out the scope of developing an application that supports predictive analytics for demand forecasting or AI-based customer support.

Scalability and Expansion

Describe the plans to scale the Hydro Mart application either for a larger user base or geographical expansion.

Describe technical and infrastructural strategies related to scaling: cloud resources, database growth, UI optimization, and so on.

Long-Term Vision

Present what the long-term vision for this **Hydro Mart** project is: either licensing to water suppliers, integration into smart city initiatives, or expanding into other delivery services.

Outline how constant innovation and user feedback are going to be the driving forces of development in the application's future.

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